FULL MOLECULAR HAMILTONIAN

From Classical to Quantum

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Contents

1 Classical Cartesian Hamiltonian

We have N_{nucl} nuclei labeled by j with masses M_j, charges $Z_j e$, positions $\mathbf{R}_j$, and conjugate momenta $\mathbf{P}_j$; and N_{elec} electrons labeled by i with mass m_e, charge $-e$, positions $\mathbf{r}_i$, and momenta $\mathbf{p}_i$. All vectors live in a space-fixed (SF) lab frame. Using Gaussian units and writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$H = \underbrace{\sum_{j=1}^{N_{\text{nucl}}} \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_{i=1}^{N_{\text{elec}}} \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \tag{1.1}$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{i\neq j}\frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j}\frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}}\right). \tag{1.2}$$

Canonical Poisson brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk}\delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij}\delta_{\alpha\beta}$.

2 Transformation to the Nuclear Center-of-Mass

2.1 New coordinates

Define the nuclear COM, $3(N-1)$ internal nuclear vectors, and electron positions in the nuclear COM frame:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j, \qquad M_N = \sum_j M_j, \tag{2.1}$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \tag{2.2}$$

The $\mathbf{R}'_j$ are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \tag{2.3}$$

2.2 Canonical momenta

This is a point transformation $\mathbf{Q}'_{\text{new}} = \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$ with $\mathbf{q}_{\text{old}} = (\mathbf{R}_j, \mathbf{r}_i)$ and $\mathbf{Q}'_{\text{new}} = (\mathbf{R}_{\text{cm}}, \mathbf{R}'_j, \mathbf{r}'_i)$. Use the generating function: $F_2 = F_2(\mathbf{q}_{\text{old}}, \mathbf{P}, t) = \mathbf{P}\cdot\mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$

$$F_2 = \mathbf{P}_{\text{cm}}\cdot\mathbf{R}_{\text{cm}}(\{\mathbf{R}_j\}) + \sum_j \mathbf{P}'_j\cdot\mathbf{R}'_j(\{\mathbf{R}_j\}) + \sum_i \mathbf{p}'_i\cdot\mathbf{r}'_i(\mathbf{R}_j, \mathbf{r}_i). \tag{2.4}$$

Then $\mathbf{P}_j = \partial F_2/\partial\mathbf{R}_j$, $\mathbf{p}_i = \partial F_2/\partial\mathbf{r}_i$. Using

$$\frac{\partial\mathbf{R}_{\text{cm}}}{\partial\mathbf{R}_j} = \frac{M_j}{M_N}\mathbb{1}, \quad \frac{\partial\mathbf{R}'_k}{\partial\mathbf{R}_j} = \left(\delta_{jk} - \tfrac{M_j}{M_N}\right)\mathbb{1}, \quad \frac{\partial\mathbf{r}'_i}{\partial\mathbf{R}_j} = -\tfrac{M_j}{M_N}\mathbb{1}, \quad \frac{\partial\mathbf{r}'_i}{\partial\mathbf{r}_j} = \delta_{ij}\mathbb{1}, \tag{2.5}$$

one obtains

$$\mathbf{p}_i = \mathbf{p}'_i, \qquad \mathbf{P}_j = \frac{M_j}{M_N}\left(\mathbf{P}_{\text{cm}} - \textstyle\sum_i \mathbf{p}'_i - \sum_k \mathbf{P}'_k\right) + \mathbf{P}'_j. \tag{2.6}$$

The constraint dual to (2.3) is $\sum_k \mathbf{P}'_k = 0$, so

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N}\big(\mathbf{P}_{\mathrm{cm}} - \textstyle\sum_i \mathbf{p}_i\big) + \mathbf{P}'_j, \qquad \mathbf{p}_i = \mathbf{p}'_i} \tag{2.7}$$

Total momentum: $\sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i = \mathbf{P}_{\mathrm{cm}}$. So $\mathbf{P}_{\mathrm{cm}}$ is the *total* (nuclei+electrons) momentum.

2.3 Hamiltonian after the transformation

Insert (2.7) into T_N in (1.1):

$$T_N = \sum_j \frac{\mathbf{P}_j^2}{2M_j} = \sum_j \frac{1}{2M_j}\left[\frac{M_j}{M_N}\big(\mathbf{P}_{\mathrm{cm}} - \textstyle\sum_i \mathbf{p}_i\big) + \mathbf{P}'_j\right]^2. \tag{2.8}$$

Expanding the square and summing,

$$T_N = \frac{(\mathbf{P}_{\mathrm{cm}} - \sum_i \mathbf{p}_i)^2}{2M_N} + \frac{1}{M_N}\big(\mathbf{P}_{\mathrm{cm}} - \sum_i \mathbf{p}_i\big)\cdot \underbrace{\sum_j \mathbf{P}'_j}_{=0} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j}. \tag{2.9}$$

Combining with T_e and V:

$$H = \frac{\mathbf{P}_{\mathrm{cm}}^2}{2M_N} \underbrace{-\frac{\mathbf{P}_{\mathrm{cm}}\cdot\sum_i \mathbf{p}_i}{M_N}}_{\text{artifact}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass-polarization}} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V(\mathbf{R}'_j, \mathbf{r}'_i). \tag{2.10}$$

Because H is independent of $\mathbf{R}_{\mathrm{cm}}$, $\mathbf{P}_{\mathrm{cm}}$ is conserved; pick the eigenvalue $\mathbf{P}_{\mathrm{cm}} = 0$ (center-of-mass at rest) and drop the trivial $\mathbf{P}_{\mathrm{cm}}$-pieces:

$$\boxed{H_{\mathrm{int}} = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + \underbrace{\frac{1}{2M_N}\Big(\sum_i \mathbf{p}_i\Big)^2}_{\text{electron mass polarization}} + V(\mathbf{R}'_j, \mathbf{r}'_i).} \tag{2.11}$$

The potential is unchanged in form because $\mathbf{R}'_j - \mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_j$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_j = \mathbf{r}_i - \mathbf{r}_j$. The mass-polarization term, $(M_N)^{-1}\sum_{i<j}\mathbf{p}_i\cdot\mathbf{p}_j + (2M_N)^{-1}\sum_i \mathbf{p}_i^2$, is the price of referring electrons to the nuclear (rather than the total) COM. This rest-frame choice is not essential: Section 2.4 shows that the *same* internal Hamiltonian follows for any $\mathbf{P}_{\mathrm{cm}}$ from a classical canonical transformation, and it is that frame-independent form we carry forward.

2.4 Eliminating the Cross-Coupling via a Classical Canonical Transformation

The rest-frame choice $\mathbf{P}_{\mathrm{cm}} = 0$ made in Section 2.3 is convenient but stronger than necessary, and we now lift it. For $\mathbf{P}_{\mathrm{cm}} \neq 0$ the Hamiltonian (2.10) carries two artifacts inherited from having referred electrons to the *nuclear* (rather than the *total*) center of mass:

$$H = \underbrace{\frac{\mathbf{P}_{\rm cm}^2}{2M_N}}_{(A)\text{ wrong total mass}} \underbrace{-\frac{\mathbf{P}_{\rm cm}\cdot\sum_i \mathbf{p}'_i}{M_N}}_{(B)\text{ cross-coupling}} + \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} + \sum_j \frac{\mathbf{P}_j^{\prime\,2}}{2M_j} + \sum_i \frac{\mathbf{p}_i^{\prime\,2}}{2m_e} + V. \tag{2.12}$$

The wrong-mass term (A) has denominator M_N instead of the proper $M_{\rm tot} = M_N + N_{\rm elec} m_e$, and (B) couples the COM motion to the internal electronic momenta. We eliminate both *exactly, for any* $\mathbf{P}_{\rm cm}$, by a single classical canonical transformation generated on phase space — so that the internal Hamiltonian carried into the body-fixed treatment of Section 3 is the frame-independent object, not a rest-frame special case. This transformation is the classical preimage of the Galilean-boost unitary that one would apply in the quantum theory.

2.4.1 Generator

Define on phase space the function

$$G \;\equiv\; \frac{m_e}{M_{\rm tot}}\,\mathbf{P}_{\rm cm}\cdot\sum_i \mathbf{r}'_i, \qquad M_{\rm tot} = M_N + N_{\rm elec} m_e. \tag{2.13}$$

G depends only on $\mathbf{P}_{\rm cm}$ and $\{\mathbf{r}'_i\}$; in particular it is independent of $\mathbf{R}_{\rm cm}$, $\mathbf{R}'_j$, $\mathbf{P}'_j$, and $\mathbf{p}'_i$.

2.4.2 Induced canonical transformation

The infinitesimal canonical transformation generated by G acts on any phase-space function f by $\delta f = \{f, G\}$. Because G is linear in $\mathbf{P}_{\rm cm}$ and $\{\mathbf{r}'_i\}$, the bracket $\{\cdot, G\}$ applied twice annihilates the canonical variables, so the finite transformation at unit parameter is given by a single application:

$$\tilde{f} = f + \{f, G\}. \tag{2.14}$$

Using $\{f, g\} = \sum(\partial_q f\,\partial_p g - \partial_p f\,\partial_q g)$, compute the action on each canonical pair.
Electron pair $(\mathbf{r}'_i, \mathbf{p}'_i)$:

$$\begin{aligned}
\{r'_{i,a}, G\} &= \tfrac{\partial G}{\partial p'_{i,a}} = 0,\\
\{p'_{i,a}, G\} &= -\tfrac{\partial G}{\partial r'_{i,a}} = -\tfrac{m_e}{M_{\rm tot}} P_{{\rm cm},a}.
\end{aligned}$$

Nuclear-COM pair $(\mathbf{R}_{\rm cm}, \mathbf{P}_{\rm cm})$:

$$\begin{aligned}
\{R_{{\rm cm},a}, G\} &= \tfrac{\partial G}{\partial P_{{\rm cm},a}} = \tfrac{m_e}{M_{\rm tot}} \sum_i r'_{i,a},\\
\{P_{{\rm cm},a}, G\} &= -\tfrac{\partial G}{\partial R_{{\rm cm},a}} = 0.
\end{aligned}$$

Internal nuclear pair $(\mathbf{R}'_j, \mathbf{P}'_j)$: G depends on neither, so $\{\mathbf{R}'_j, G\} = \{\mathbf{P}'_j, G\} = 0$. Hence the finite canonical transformation reads

$$\boxed{\begin{aligned}
\tilde{\mathbf{r}}'_i &= \mathbf{r}'_i, & \tilde{\mathbf{p}}'_i &= \mathbf{p}'_i - \tfrac{m_e}{M_{\rm tot}}\mathbf{P}_{\rm cm},\\
\tilde{\mathbf{R}}_{\rm cm} &= \mathbf{R}_{\rm cm} + \tfrac{m_e}{M_{\rm tot}}\textstyle\sum_i \mathbf{r}'_i, & \tilde{\mathbf{P}}_{\rm cm} &= \mathbf{P}_{\rm cm},\\
\tilde{\mathbf{R}}'_j &= \mathbf{R}'_j, & \tilde{\mathbf{P}}'_j &= \mathbf{P}'_j.
\end{aligned}} \tag{2.15}$$

2.4.3 Canonicity check

The only non-obvious Poisson bracket is between the two simultaneously shifted variables:

$$\begin{aligned}\{\tilde{p}'_{i,a}, \tilde{R}_{\mathrm{cm},b}\} &= \left\{\mathbf{p}'_{i,a} - \tfrac{m_e}{M_{\mathrm{tot}}} P_{\mathrm{cm},a},\ R_{\mathrm{cm},b} + \tfrac{m_e}{M_{\mathrm{tot}}} \textstyle\sum_l r'_{l,b}\right\} \\ &= \underbrace{\{p'_{i,a}, R_{\mathrm{cm},b}\}}_{0} + \tfrac{m_e}{M_{\mathrm{tot}}} \sum_l \underbrace{\{p'_{i,a}, r'_{l,b}\}}_{-\delta_{il}\delta_{ab}} \\ &\quad - \tfrac{m_e}{M_{\mathrm{tot}}} \underbrace{\{P_{\mathrm{cm},a}, R_{\mathrm{cm},b}\}}_{-\delta_{ab}} - \tfrac{m_e^2}{M_{\mathrm{tot}}^2} \sum_l \underbrace{\{P_{\mathrm{cm},a}, r'_{l,b}\}}_{0} \\ &= -\tfrac{m_e}{M_{\mathrm{tot}}} \delta_{ab} + \tfrac{m_e}{M_{\mathrm{tot}}} \delta_{ab} = 0. \end{aligned} \tag{2.16}$$

All other elementary brackets are preserved by inspection (each is either unchanged or trivially zero), so (2.15) is canonical.

2.4.4 Transformed Hamiltonian

Invert (2.15) for substitution into (2.12): $\mathbf{p}'_i = \tilde{\mathbf{p}}'_i + (m_e/M_{\mathrm{tot}})\mathbf{P}_{\mathrm{cm}}$. The potential V depends only on $\mathbf{R}'_j$ and $\mathbf{r}'_i$, which are unchanged, so V is form-invariant. Working term by term.

Cross-coupling (B):

$$\begin{aligned} -\frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N} &= -\frac{\mathbf{P}_{\mathrm{cm}} \cdot \left(\sum_i \tilde{\mathbf{p}}'_i + N_{\mathrm{elec}} \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}}\right)}{M_N} \\ &= -\frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N} - \frac{N_{\mathrm{elec}} m_e \, \mathbf{P}_{\mathrm{cm}}^2}{M_N M_{\mathrm{tot}}}. \end{aligned} \tag{2.17}$$

Mass-polarization quadratic:

$$\begin{aligned} \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} &= \frac{\left(\sum_i \tilde{\mathbf{p}}'_i + N_{\mathrm{elec}} \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}}\right)^2}{2M_N} \\ &= \frac{(\sum_i \tilde{\mathbf{p}}'_i)^2}{2M_N} + \frac{N_{\mathrm{elec}} m_e \, \mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N M_{\mathrm{tot}}} + \frac{N_{\mathrm{elec}}^2 m_e^2 \, \mathbf{P}_{\mathrm{cm}}^2}{2 M_N M_{\mathrm{tot}}^2}. \end{aligned} \tag{2.18}$$

Electron kinetic energy:

$$\begin{aligned} \sum_i \frac{\mathbf{p}_i'^{\,2}}{2m_e} &= \sum_i \frac{(\tilde{\mathbf{p}}'_i + \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}})^2}{2m_e} \\ &= \sum_i \frac{\tilde{\mathbf{p}}_i'^{\,2}}{2m_e} + \frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_{\mathrm{tot}}} + \frac{N_{\mathrm{elec}} m_e \, \mathbf{P}_{\mathrm{cm}}^2}{2 M_{\mathrm{tot}}^2}. \end{aligned} \tag{2.19}$$

Coefficient of $\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i$: Summing (2.17), (2.18), (2.19),

$$-\frac{1}{M_N} + \frac{N_{\mathrm{elec}} m_e}{M_N M_{\mathrm{tot}}} + \frac{1}{M_{\mathrm{tot}}} = \frac{-M_{\mathrm{tot}} + N_{\mathrm{elec}} m_e + M_N}{M_N M_{\mathrm{tot}}} = 0, \tag{2.20}$$

using $M_{\mathrm{tot}} = M_N + N_{\mathrm{elec}} m_e$. *The cross-coupling (B) is killed identically.*

Coefficient of $\mathbf{P}_{\mathrm{cm}}^2$: Adding the $\mathbf{P}_{\mathrm{cm}}^2/(2M_N)$ from (A) plus the residual $\mathbf{P}_{\mathrm{cm}}^2$-pieces from

(2.17)–(2.19),

$$\begin{aligned}\frac{1}{2M_N}-\frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}}+\frac{N_{\text{elec}}^2 m_e^2}{2M_N M_{\text{tot}}^2}+\frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} &= \frac{1}{2M_N}-\frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}}+\frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2}\left(\frac{N_{\text{elec}}m_e}{M_N}+1\right)\\ &= \frac{1}{2M_N}-\frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}}+\frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2}\cdot\frac{M_{\text{tot}}}{M_N}\\ &= \frac{1}{2M_N}-\frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}}+\frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}}\\ &= \frac{1}{2M_N}-\frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}}=\frac{M_{\text{tot}}-N_{\text{elec}}m_e}{2M_N M_{\text{tot}}}=\frac{M_N}{2M_N M_{\text{tot}}}=\frac{1}{2M_{\text{tot}}}.\end{aligned} \tag{2.21}$$

The wrong-mass (A) is corrected to M_{tot}.

Assembling everything,

$$\boxed{\tilde{H}=\frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}}\;+\;\sum_j\frac{\mathbf{P}_j'^{\,2}}{2M_j}\;+\;\sum_i\frac{\tilde{\mathbf{p}}_i'^{\,2}}{2m_e}\;+\;\frac{1}{2M_N}\Big(\sum_i\tilde{\mathbf{p}}_i'\Big)^2\;+\;V(\mathbf{R}_j',\mathbf{r}_i').} \tag{2.22}$$

The COM kinetic energy carries the *correct* total mass M_{tot}. The internal piece is structurally identical to (2.11): the same mass-polarization $(\sum_i\tilde{\mathbf{p}}_i')^2/(2M_N)$, the same nuclear kinetic energy, the same potential. Since $\tilde{H}$ is independent of $\tilde{\mathbf{R}}_{\text{cm}}$, $\tilde{\mathbf{P}}_{\text{cm}}=\mathbf{P}_{\text{cm}}$ is conserved, and one may proceed with the internal dynamics for *any* value of $\mathbf{P}_{\text{cm}}$ — not just zero — with no kinetic artifact.

2.4.5 Geometric interpretation

The new "COM coordinate" is

$$\tilde{\mathbf{R}}_{\text{cm}}=\mathbf{R}_{\text{cm}}+\frac{m_e}{M_{\text{tot}}}\sum_i\mathbf{r}_i'=\frac{M_N\mathbf{R}_{\text{cm}}+m_e\sum_i\mathbf{r}_i}{M_{\text{tot}}}=\mathbf{R}_{\text{tot}}^{\text{CoM}}, \tag{2.23}$$

using $\mathbf{r}_i=\mathbf{R}_{\text{cm}}+\mathbf{r}_i'$. The canonical transformation has *quietly relabeled* the COM coordinate from nuclear to total COM, while *leaving the relative coordinates $\mathbf{R}_j'$ and $\mathbf{r}_i'$ untouched*. This is precisely what makes the framework still "nuclear-COM-based": the Born–Oppenheimer-friendly relative coordinates of Section 2.1 survive unchanged. The momentum shift $\tilde{\mathbf{p}}_i'=\mathbf{p}_i'-(m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$ is the kinetic compensation required to keep the Poisson brackets canonical: in Galilean terms, each electron's lab-frame momentum is reduced by $m_e\mathbf{V}_{\text{CoM}}$ with $\mathbf{V}_{\text{CoM}}=\mathbf{P}_{\text{cm}}/M_{\text{tot}}$.

The canonical transformation has thus cleanly separated the total COM motion *exactly and for any* $\mathbf{P}_{\text{cm}}$: in (2.22) the COM appears only through the decoupled free term $\mathbf{P}_{\text{cm}}^2/2M_{\text{tot}}$ — carrying the *correct* total mass M_{tot} and free of the cross-coupling (B) — while every other term depends only on the internal variables. It is this total Hamiltonian (2.22), exact for *any* $\mathbf{P}_{\text{cm}}$ and not merely the rest frame, that we carry into the body-fixed treatment of Section 3.

3 BF frame for both nuclei and electrons

3.1 Setup

With Eckart's frame [1] fixed by the *nuclei*,

$$\mathbf{R}'_j = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{R}}_j(Q), \qquad \mathbf{r}'_i = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{r}}_i. \tag{3.1}$$

The Eckart conditions $\sum_j M_j \bar{\mathbf{R}}_j = 0$ and $\sum_j M_j \bar{\mathbf{R}}_j^{\rm eq} \times \bar{\mathbf{R}}_j = 0$ fix the orientation of $\mathbf{S}$ from the *nuclear* positions; the same $\mathbf{S}$ is then used to define BF electron coordinates $\bar{\mathbf{r}}_i$. The nuclear shape is parametrized by rectilinear mass-weighted **normal coordinates** Q_b,

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\rm eq} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb}\, Q_b, \tag{3.1a}$$

with mass-weighted normal-mode vectors $\mathbf{l}_{jb}$ obeying orthonormality and the Eckart (Sayvetz) conditions [1, 2]

$$\sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}, \qquad \sum_j \sqrt{M_j}\,\mathbf{l}_{jb} = 0, \qquad \sum_j \sqrt{M_j}\,\bar{\mathbf{R}}_j^{\rm eq}\times\mathbf{l}_{jb} = 0, \tag{3.1b}$$

so that $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ is constant.
BF angular velocity:

$$(\mathbf{S}^T\dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\,\omega_\gamma \Longrightarrow \mathbf{S}^T\dot{\mathbf{S}}\,\mathbf{v} = \boldsymbol{\omega}\times\mathbf{v}. \tag{3.2}$$

Velocities (using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$):

$$\dot{\mathbf{R}}'_j = \mathbf{S}\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big), \qquad \dot{\mathbf{r}}'_i = \mathbf{S}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big). \tag{3.3}$$

3.2 From the internal Hamiltonian to a working Routhian

I need an action principle for the body-fixed *nuclear* coordinates. The electrons, however, are best left in the canonical (momentum) form they already carry in (2.22): their kinetic energy and mass polarization form a rotational scalar that passes through the body-fixed transformation unchanged. I therefore Legendre-transform *only* the COM and nuclear momenta back to velocities — keeping $\mathbf{P}_{\rm cm}$ arbitrary, never setting it to zero — and keep the electron momenta $\tilde{\mathbf{p}}'_i$ as canonical variables. The relevant Hamilton equations from (2.22) are

$$\dot{\bar{\mathbf{R}}}_{\rm cm} = \mathbf{P}_{\rm cm}/M_{\rm tot}, \qquad \dot{\mathbf{R}}'_j = \mathbf{P}'_j/M_j, \qquad \dot{\mathbf{r}}'_i = \tilde{\mathbf{p}}'_i/m_e + (\textstyle\sum_l \tilde{\mathbf{p}}'_l)/M_N, \tag{3.4}$$

the last of which inverts to the electron momentum–velocity relation [recorded for the velocity-form cross-check below, around (3.18)]

$$\tilde{\mathbf{p}}'_i = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\rm tot}}\sum_l \dot{\mathbf{r}}'_l, \qquad \sum_l \tilde{\mathbf{p}}'_l = \frac{m_e M_N}{M_{\rm tot}}\sum_i \dot{\mathbf{r}}'_i. \tag{3.5}$$

Legendre-transforming the COM and nuclear momenta only, $\mathcal{R} = \mathbf{P}_{\rm cm}\cdot\dot{\tilde{\mathbf{R}}}_{\rm cm} + \sum_j \mathbf{P}'_j\cdot\dot{\mathbf{R}}'_j - \tilde{H}$, yields the **Routhian**

$$\boxed{\mathcal{R} = \tfrac{1}{2}M_{\rm tot}|\dot{\tilde{\mathbf{R}}}_{\rm cm}|^2 + \tfrac{1}{2}\sum_j M_j|\dot{\mathbf{R}}'_j|^2 - H_e(\tilde{\mathbf{p}}'_i) - V, \qquad H_e \equiv \sum_i \frac{\tilde{\mathbf{p}}_i'^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \tilde{\mathbf{p}}'_i\Big)^2,} \tag{3.6}$$

which is a Lagrangian in $(\tilde{\mathbf{R}}_{\rm cm}, \mathbf{R}'_j)$ and *minus* a Hamiltonian in $(\mathbf{r}'_i, \tilde{\mathbf{p}}'_i)$: extremizing $\int \mathcal{R}\,dt$ gives Euler–Lagrange equations for the COM and nuclei and Hamilton's equations for the electrons. The electronic kinetic energy and mass polarization thus retain their (2.22) momentum form H_e — they are never converted to velocity form. (The overall minus sign on H_e is the Routhian's partial-Legendre signature; the mass polarization keeps its physical positive coefficient $+1/2M_N$ inside H_e, and the whole H_e re-enters the Hamiltonian with a plus sign when it is reassembled in (3.22).)

3.3 Body-fixed Routhian

Substitute the body-fixed velocities (3.3) into the nuclear term of (3.6); the COM term is untouched by the rotation. The electron Hamiltonian H_e is a *rotational scalar* — it is built from the dot products $\tilde{\mathbf{p}}'_i\cdot\tilde{\mathbf{p}}'_l$ — so under the passive rotation to the body frame it is unchanged in form. Defining the body-fixed electron momenta $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T\tilde{\mathbf{p}}'_i$ (a proper rotation, so $|\bar{\mathbf{p}}_i|^2 = |\tilde{\mathbf{p}}'_i|^2$ and $\sum_i \bar{\mathbf{p}}_i = \mathbf{S}^T\sum_i \tilde{\mathbf{p}}'_i$, whence $(\sum_i \bar{\mathbf{p}}_i)^2 = (\sum_i \tilde{\mathbf{p}}'_i)^2$),

$$H_e = \sum_i \frac{\tilde{\mathbf{p}}_i'^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \tilde{\mathbf{p}}'_i\Big)^2 = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2. \tag{3.6a}$$

Defining $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$, the body-fixed Routhian is therefore

$$\mathcal{R} = \tfrac{1}{2}M_{\rm tot}|\dot{\tilde{\mathbf{R}}}_{\rm cm}|^2 + \tfrac{1}{2}\sum_j M_j|\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 - V, \tag{3.7}$$

in which only the nuclear kinetic term carries the body-fixed angular velocity $\boldsymbol{\omega}$. The potential is now manifestly $\mathbf{S}$-independent:

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{j\neq k}\frac{Z_jZ_ke^2}{\bar{R}_{jk}} - \sum_{j,i}\frac{Z_je^2}{|\bar{\mathbf{R}}_j - \bar{\mathbf{r}}_i|} + \frac{1}{2}\sum_{i\neq l}\frac{e^2}{\bar{r}_{il}}\right). \tag{3.8}$$

Rotational symmetry is rendered explicit; the BF electronic problem at fixed Q is self-contained.

3.4 Expanding (3.7)

3.4.1 Generic squared-term identity

For any vectors $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega} \in \mathbb{R}^3$, the BCH-style identity $\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a}) = \boldsymbol{\omega}\,a^2 - \mathbf{a}(\mathbf{a}\cdot\boldsymbol{\omega})$ gives

$$\begin{aligned}|\boldsymbol{\omega}\times\mathbf{a}|^2 = (\boldsymbol{\omega}\times\mathbf{a})\cdot(\boldsymbol{\omega}\times\mathbf{a}) \;&=\; \boldsymbol{\omega}\cdot\big[\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a})\big]\\ = \boldsymbol{\omega}\cdot\big[\boldsymbol{\omega}\,a^2 - \mathbf{a}(\mathbf{a}\cdot\boldsymbol{\omega})\big] \;&=\; \boldsymbol{\omega}^T(a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\,\boldsymbol{\omega},\end{aligned} \tag{3.9a}$$

while the scalar triple product gives

$$(\boldsymbol{\omega}\times\mathbf{a})\cdot\dot{\mathbf{a}} \;=\; \boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}}). \tag{3.9b}$$

Hence, for any constant mass-like coefficient m,

$$\begin{aligned}\tfrac{m}{2}\big|\boldsymbol{\omega}\times\mathbf{a}+\dot{\mathbf{a}}\big|^2 &= \tfrac{m}{2}\big[|\boldsymbol{\omega}\times\mathbf{a}|^2+2(\boldsymbol{\omega}\times\mathbf{a})\cdot\dot{\mathbf{a}}+|\dot{\mathbf{a}}|^2\big]\\ &= \underbrace{\tfrac{m}{2}\,\boldsymbol{\omega}^T(a^2\mathbb{1}-\mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m\,\boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}})}_{\text{Coriolis-type}} + \underbrace{\tfrac{m}{2}|\dot{\mathbf{a}}|^2}_{\text{vibrational}}\;.\end{aligned} \tag{3.9c}$$

3.4.2 Application to the nuclear term

Only the nuclear kinetic term of the Routhian (3.7) carries $\boldsymbol{\omega}$. Apply (3.9c) with $m\to M_j$, $\mathbf{a}\to\bar{\mathbf{R}}_j$ and sum over j:

$$\tfrac{1}{2}\sum_j M_j\big|\boldsymbol{\omega}\times\bar{\mathbf{R}}_j+\dot{\bar{\mathbf{R}}}_j\big|^2 \;=\; \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_N\boldsymbol{\omega} \;+\; \boldsymbol{\omega}\cdot\mathbf{L}_N^{\mathrm{vib}} \;+\; T_{\mathrm{vib}}^N. \tag{3.9d}$$

The electronic kinetic energy is *not* expanded this way — it is carried as the rotational scalar H_e of (3.6a). Only for the velocity-form cross-check of § 3.6 do we record the decomposition that the electronic and mass-polarization terms *would* produce if the electrons were instead kept in velocity form,

$$\tfrac{m_e}{2}\sum_i\big|\boldsymbol{\omega}\times\bar{\mathbf{r}}_i+\dot{\bar{\mathbf{r}}}_i\big|^2 = \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_e\boldsymbol{\omega}+\boldsymbol{\omega}\cdot\mathbf{L}_e+T_e^{\mathrm{BF}}, \qquad -\tfrac{m_e^2}{2M_{\mathrm{tot}}}\big|\boldsymbol{\omega}\times\bar{\mathbf{r}}+\dot{\bar{\mathbf{r}}}\big|^2 = -\tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_X\boldsymbol{\omega}-\boldsymbol{\omega}\cdot\mathbf{L}_X-T_X^{\mathrm{mp}}. \tag{3.9e}$$

The constituent inertia tensors are

$$\mathbf{I}_N = \sum_j M_j(\bar{R}_j^2\mathbb{1}-\bar{\mathbf{R}}_j\bar{\mathbf{R}}_j^T), \qquad \mathbf{I}_e = m_e\sum_i(\bar{r}_i^2\mathbb{1}-\bar{\mathbf{r}}_i\bar{\mathbf{r}}_i^T), \tag{3.9}$$

$$\mathbf{I}_X = \tfrac{m_e^2}{M_{\mathrm{tot}}}(\bar{r}^2\mathbb{1}-\bar{\mathbf{r}}\bar{\mathbf{r}}^T), \tag{3.10}$$

the angular-momentum-like (velocity-form) quantities are

$$\mathbf{L}_N^{\mathrm{vib}} = \sum_j M_j\bar{\mathbf{R}}_j\times\dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e\sum_i\bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \tfrac{m_e^2}{M_{\mathrm{tot}}}\bar{\mathbf{r}}\times\dot{\bar{\mathbf{r}}}, \tag{3.11}$$

and the "vibrational" kinetic pieces are

$$T_{\mathrm{vib}}^N = \tfrac{1}{2}\sum_j M_j|\dot{\bar{\mathbf{R}}}_j|^2 = \tfrac{1}{2}\sum_b\dot{Q}_b^2, \quad T_e^{\mathrm{BF}} = \tfrac{m_e}{2}\sum_i|\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\mathrm{mp}} = \tfrac{m_e^2}{2M_{\mathrm{tot}}}|\dot{\bar{\mathbf{r}}}|^2. \tag{3.12}$$

3.4.3 Assembled Routhian

Inserting the nuclear expansion (3.9d) into (3.7), together with the decoupled COM term, the electron scalar H_e, and the potential:

$$\boxed{\mathcal{R} = \tfrac{1}{2}M_{\mathrm{tot}}|\dot{\bar{\mathbf{R}}}_{\mathrm{cm}}|^2 + \underbrace{\tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_N\,\boldsymbol{\omega}+\boldsymbol{\omega}\cdot\mathbf{L}_N^{\mathrm{vib}}+T_{\mathrm{vib}}^N}_{\text{nuclear, velocity form}} - H_e \;-\; V.} \tag{3.13}$$

The rotational kernel is the **nuclear** inertia $\mathbf{I}_N$ *alone*: because the electrons are carried as the scalar H_e rather than expanded in $\boldsymbol{\omega}$, no electronic or mass-polarization inertia ($\mathbf{I}_e$, $\mathbf{I}_X$) enters the kernel. The electrons couple to rotation only *kinematically*, through their angular momentum, which we identify next. The body-fixed electron momenta and the **electronic angular momentum** are

$$\bar{\mathbf{p}}_i = \mathbf{S}^T \tilde{\mathbf{p}}'_i, \qquad \mathbf{L}_{\text{elec}} \equiv \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i. \tag{3.14}$$

Remark. The electronic and mass-polarization inertia/angular-momentum quantities $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ of (3.9)–(3.11), and the velocity-form decomposition (3.9e), are needed *only* in the cross-check of § 3.6; they play no role in the Routhian (3.13) itself.

3.5 Canonical momenta and the total angular momentum

Compute the conjugate momenta from the Routhian $\mathcal{R}$. The cyclic COM coordinate gives the conserved total momentum $\mathbf{P}_{\text{cm}} = \partial\mathcal{R}/\partial\dot{\bar{\mathbf{R}}}_{\text{cm}} = M_{\text{tot}}\dot{\bar{\mathbf{R}}}_{\text{cm}}$; being decoupled, it contributes nothing to the internal momenta below. The electron momenta $\bar{\mathbf{p}}_i$ are already canonical [eq. (3.14)]; it remains to find the rotational and vibrational momenta, most transparently by differentiating the unexpanded nuclear term of (3.7).

3.5.1 Nuclear angular momentum $\mathbf{J}_N = \partial\mathcal{R}/\partial\boldsymbol{\omega}$

In the Routhian (3.13) only the nuclear rotational kernel and the nuclear Coriolis piece carry $\boldsymbol{\omega}$ — the electron scalar H_e and the COM term do not. The momentum conjugate to the body-fixed angular velocity is therefore the *nuclear* angular momentum

$$\mathbf{J}_N \;\equiv\; \frac{\partial\mathcal{R}}{\partial\boldsymbol{\omega}} = \mathbf{I}_N\,\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}. \tag{3.15}$$

3.5.2 Total angular momentum $\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}}$

The body-fixed frame $\mathbf{S}(\Omega)$ rotates the nuclei *and* the electrons together [eq. (3.1)], so the total angular momentum conjugate to the Euler angles — the physical, conserved rovibronic angular momentum — is the sum of the nuclear and electronic contributions. The electronic part is the canonical electron angular momentum $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ of (3.14), the body-fixed image of $\sum_i \mathbf{r}'_i \times \tilde{\mathbf{p}}'_i$ (the cross product co-rotates with $\mathbf{S}$). Hence

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \tag{3.16}$$

3.5.3 Vibrational momentum $P_b = \partial\mathcal{R}/\partial\dot{Q}_b$

Only $\dot{\bar{\mathbf{R}}}_j$ carries $\dot{Q}_b$ dependence; with $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a),

$$P_b \;\equiv\; \frac{\partial\mathcal{R}}{\partial\dot{Q}_b} = \sum_j M_j\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot\frac{\partial\bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j}\,\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot\mathbf{l}_{jb}. \tag{3.17}$$

Separating the rotational and vibrational pieces defines the **Coriolis vectors $\boldsymbol{\sigma}_b$**,

$$\boldsymbol{\sigma}_b \equiv \sum_j M_j\, \bar{\mathbf{R}}_j \times \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j}\, \bar{\mathbf{R}}_j \times \mathbf{l}_{jb} = \sum_a \boldsymbol{\zeta}_{ab}\, Q_a, \qquad \boldsymbol{\zeta}_{ab} \equiv \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}, \tag{3.17a}$$

the second form following from (3.1a) with the Eckart condition $\sum_j \sqrt{M_j}\bar{\mathbf{R}}_j^{\mathrm{eq}} \times \mathbf{l}_{jb} = 0$. The vibrational metric is the identity, $\sum_j M_j(\partial\bar{\mathbf{R}}_j/\partial Q_a)\cdot(\partial\bar{\mathbf{R}}_j/\partial Q_b) = \sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$, so the canonical vibrational momentum carries *no* Wilson G-matrix,

$$P_b \;=\; \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b(Q) \;+\; \dot{Q}_b. \tag{3.17b}$$

3.6 Key identity

3.6.1 The clean split

The total-angular-momentum decomposition (3.16) *is* the key identity: subtracting the electronic part from both sides,

$$\boxed{\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N(Q)\,\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}.} \tag{3.19}$$

The right side is a purely nuclear-velocity quantity, with the *nuclear* inertia $\mathbf{I}_N$ as rotational kernel. The entire electronic and mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\mathrm{elec}}$; it never enters the kernel, because in the Routhian (3.13) the electrons were carried as the scalar H_e and contributed no inertia at all.

3.6.2 Cross-check: the $\mathbf{I}_e/\mathbf{I}_X$ cancellation of the velocity-form route

Had we instead converted the electrons to velocity form — expanding their kinetic energy via (3.9e) — the electronic inertia and Coriolis terms would have entered *both* $\mathbf{J}$ and $\mathbf{L}_{\mathrm{elec}}$, and (3.19) would emerge only after an exact cancellation. The velocity-form electron momentum is $\bar{\mathbf{p}}_i = m_e(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\mathrm{tot}}}(\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}})$ [the body-fixed image of (3.5)]; substituting into $\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ and using the BAC–CAB identity $\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a}) = (a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}$ on the two $\boldsymbol{\omega}$-terms collapses the four pieces to

$$\mathbf{L}_{\mathrm{elec}} = (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X), \tag{3.18}$$

with $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ from (3.9)–(3.11). The velocity-form total angular momentum would read $\mathbf{J} = (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_e - \mathbf{L}_X)$ [from (3.9d)+(3.9e)], so

$$\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}} + \underbrace{\big[(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)\big] - \mathbf{L}_{\mathrm{elec}}}_{=0 \text{ by } (3.18)} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}},$$

reproducing (3.19): the electronic inertia $\mathbf{I}_e$ and the mass-polarization inertia $\mathbf{I}_X$ cancel *exactly*. This "algebraic miracle" of the Eckart construction is automatic in the mixed treatment, where $\mathbf{I}_e$ and $\mathbf{I}_X$ never arise.

3.6.3 Inversion: the Watson reciprocal inertia

To invert (3.19) for $\boldsymbol{\omega}$ we must re-express the velocity-form $\mathbf{L}_N^{\mathrm{vib}}$ in canonical variables. In normal coordinates $\mathbf{L}_N^{\mathrm{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j = \sum_b \boldsymbol{\sigma}_b \, \dot{Q}_b$; eliminating $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$ from (3.17b),

$$\mathbf{L}_N^{\mathrm{vib}} = \sum_b \boldsymbol{\sigma}_b \big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\omega}, \qquad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}} \equiv \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^{\mathsf{T}}, \quad (3.20a)$$

with $\boldsymbol{\pi}$ the **field-free vibrational angular momentum**. Substituting into (3.19), $\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})\boldsymbol{\omega} + \boldsymbol{\pi} \equiv \mathbf{I}'\boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boldsymbol{\omega} = \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi}\big), \qquad \boxed{\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = \big(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\big)^{-1},} \quad (3.20)$$

the **Watson reciprocal-inertia tensor** [3]. The rotation–vibration coupling is now carried by the modified inertia $\mathbf{I}'$ and the uniform shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi}$ — with no Wilson G-matrix.

3.7 Hamiltonian in canonical variables

We now assemble the body-fixed Hamiltonian in the canonical variables $(\Omega, \mathbf{J};\, Q, P_b;\, \bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$. The total energy splits as $\tilde{H} = \mathbf{P}_{\mathrm{cm}}^2/2M_{\mathrm{tot}} + H_{\mathrm{int}}$, the COM piece being a decoupled spectator. The internal Hamiltonian is the sum of the nuclear rotation–vibration kinetic energy, the electron scalar, and the potential,

$$H_{\mathrm{int}} = T_N^{\mathrm{rot+vib}} + H_e + V, \qquad T_N^{\mathrm{rot+vib}} \equiv \tfrac{1}{2}\boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega}\cdot\mathbf{L}_N^{\mathrm{vib}} + T_{\mathrm{vib}}^N,$$

read off the Routhian (3.13). The electron scalar H_e [eq. (3.6a)] and the potential V are already canonical; only $T_N^{\mathrm{rot+vib}}$ must be re-expressed through the momenta $(\mathbf{J}, P_b)$.

3.7.1 Nuclear rotation–vibration block

$T_N^{\mathrm{rot+vib}}$ is homogeneous-quadratic in the nuclear velocities $(\boldsymbol{\omega}, \dot{Q}_b)$, so by Euler's theorem $2T_N^{\mathrm{rot+vib}} = \mathbf{J}_N\cdot\boldsymbol{\omega} + \sum_b P_b \dot{Q}_b$, with conjugate momenta $\mathbf{J}_N = \mathbf{J} - \mathbf{L}_{\mathrm{elec}}$ [eqs. (3.15),(3.19)] and P_b [eq. (3.17b)]. Substituting $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$ and $\sum_b P_b \boldsymbol{\sigma}_b = \boldsymbol{\pi}$,

$$2T_N^{\mathrm{rot+vib}} = (\mathbf{J} - \mathbf{L}_{\mathrm{elec}})\cdot\boldsymbol{\omega} + \sum_b P_b\big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = (\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi})\cdot\boldsymbol{\omega} + \sum_b P_b^2.$$

Inserting $\boldsymbol{\omega} = \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi})$ from (3.20),

$$\boxed{T_N^{\mathrm{rot+vib}}\Big|_{\mathrm{canonical}} = \tfrac{1}{2}(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}\,(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi}) + \tfrac{1}{2}\sum_b P_b^2.} \quad (3.20d)$$

The vibrational kinetic energy is the *bare* $\frac{1}{2}\sum_b P_b^2$; the entire rotation–vibration coupling is carried by $\boldsymbol{\pi}$ and by the Watson inertia $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ — no Wilson G-matrix survives.

3.7.2 Electronic block

The electronic kinetic energy and mass polarization require no further work: they are the rotational scalar H_e of (3.6a), carried unchanged from the Step-2 Hamiltonian (2.22),

$$\boxed{H_e = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2.} \tag{3.21}$$

Because H_e is built from the rotation-invariant dot products $\bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l$ and $\bar{\mathbf{p}}_i = \mathbf{S}^T\tilde{\mathbf{p}}_i'$ is a passive rotation, its body-fixed form is *identical* to the Step-2 form (2.22) under $\tilde{\mathbf{p}}_i' \to \bar{\mathbf{p}}_i$ — no Legendre transform or angular-momentum expansion is needed. The mass polarization survives the body-fixed transformation intact in functional form.

Reduced-mass check ($N_{\text{elec}} = 1$): $H_e = \bar{\mathbf{p}}_1^2/(2m_e) + \bar{\mathbf{p}}_1^2/(2M_N) = \bar{\mathbf{p}}_1^2/(2\mu_e)$ with $\mu_e = m_e M_N/(m_e + M_N) = m_e M_N/M_{\text{tot}}$. ✓

3.7.3 Final assembly: (3.22)

Add (3.20d) and (3.21), include $V = V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$ from (3.8), and use $H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V$:

$$\boxed{\begin{aligned} H_{\text{int}} &= \tfrac{1}{2}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big)^T \boldsymbol{\mu}(Q)\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big) \\ &\quad + \tfrac{1}{2}\sum_b P_b^2 \\ &\quad + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 \\ &\quad + V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \end{aligned}} \tag{3.22}$$

Restoring the decoupled COM piece, the complete canonical Hamiltonian is $\tilde{H} = \mathbf{P}_{\text{cm}}^2/2M_{\text{tot}} + H_{\text{int}}$ with H_{int} given by (3.22) — exact for *any* $\mathbf{P}_{\text{cm}}$. The COM term is a conserved free-particle spectator that decouples from every body-fixed variable, so the quantization of Section 4 proceeds with H_{int}.

Here $\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b$ is the field-free vibrational angular momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) and constants $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; the vibrational metric being the identity, no Wilson G-matrix enters.

$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ is the BF canonical electronic angular momentum, and crucially:

- $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ is the **Watson reciprocal-inertia tensor**, built from nuclear quantities alone: *no* electron or mass-polarization corrections enter the rotational kernel — they were absorbed into $\mathbf{L}_{\text{elec}}$ via (3.19).
- The electron contribution to rotation appears entirely as the Coriolis-like coupling $-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}$.
- The mass polarization is preserved in its $\big(\sum_i \bar{\mathbf{p}}_i\big)^2/(2M_N)$ form.

3.8 Complete coupling enumeration

Expanding the rotational kernel:

$$\tfrac{1}{2}\big(\mathbf{J}-\mathbf{L}_{\text{elec}}-\boldsymbol{\pi}\big)^T\boldsymbol{\mu}\big(\mathbf{J}-\mathbf{L}_{\text{elec}}-\boldsymbol{\pi}\big) =$$

$$\underbrace{\tfrac{1}{2}\mathbf{J}^T\boldsymbol{\mu}\mathbf{J}}_{\text{rotation}}\ \underbrace{-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}}_{\text{rot–vib Coriolis}}\ \underbrace{-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}}_{\text{rot–elec Coriolis}}\ \underbrace{+\boldsymbol{\pi}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}}_{\text{vib–elec angular}}+\tfrac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}+\tfrac{1}{2}\mathbf{L}_{\text{elec}}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}.$$

All couplings in (3.22), referred to the standard analyses [3, 4]:

1. **Pure rotation, anisotropic top:** $\frac{1}{2}\mathbf{J}^T\boldsymbol{\mu}(Q)\mathbf{J}$ — Q-dependent moments of inertia (rot–vib coupling through inertia).
2. **Rot–vib Coriolis** [5]: $-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}$.
3. **Rot–electron Coriolis:** $-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}$ — the kinetic source of Λ-doubling [6, 7], Renner–Teller [8], and spin–rotation [9] analogues.
4. **Vib–electron angular coupling:** $\boldsymbol{\pi}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}$ — exact non-adiabatic angular coupling intrinsic to BF embedding [10, 11, 12].
5. **Quadratic vibrational angular momentum:** $\frac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}$.
6. **Quadratic electronic angular momentum:** $\frac{1}{2}\mathbf{L}_{\text{elec}}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}$.
7. **Pure vibrational kinetic:** $\frac{1}{2}\sum_b P_b^2$.
8. **Electronic kinetic:** $\sum_i \bar{\mathbf{p}}_i^2/(2m_e)$.
9. **Mass-polarization, diagonal:** $\sum_i \bar{\mathbf{p}}_i^2/(2M_N)$ — modifies effective electron mass [10, 4].
10. **Mass-polarization, off-diagonal (two-electron):** $\sum_{i<l}\bar{\mathbf{p}}_i\cdot\bar{\mathbf{p}}_l/M_N$.
11. **Coulomb:** $V_{NN}(Q)+V_{Ne}(Q,\{\bar{\mathbf{r}}_i\})+V_{ee}(\{\bar{\mathbf{r}}_i\})$. All Euler-angle independent.

Compared to the previous (electrons-in-SF) framework: kinetic rot–electron coupling has *replaced* potential rot–electron coupling. V_{Ne} is now manifestly $\mathbf{S}$-independent — the canonical chemistry advantage.

4 Quantization

The classical internal Hamiltonian (3.22) is parametrized by curvilinear coordinates (Euler angles ϕ, θ, χ and vibrational normal coordinates Q_b) and BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$. Canonical quantization in curvilinear coordinates requires care: the naïve substitution $\pi \to -i\hbar\partial$ does not produce a Hermitian operator on the natural Hilbert space. The Podolsky (Laplace–Beltrami) prescription resolves this; the residual ordering produces the Watson pseudopotential, and the BF angular momentum acquires an *anomalous* commutator algebra.

4.1 The quantization problem in curvilinear coordinates

Label the configuration-space coordinates q^A:

$$\{q^A\} \;=\; \big\{\, \phi, \theta, \chi,\; Q_1, \ldots, Q_{3N_{\text{nucl}}-6},\; \bar{\mathbf{r}}_1, \ldots, \bar{\mathbf{r}}_{N_{\text{elec}}} \,\big\}, \tag{4.1}$$

with conjugate momenta π_A. The classical kinetic energy from (3.22) is a quadratic form

$$T \;=\; \tfrac{1}{2}\, g^{AB}(q)\, \pi_A \pi_B, \tag{4.2}$$

where g^{AB} is the inverse of the configuration-space metric g_{AB} defined by $T = \frac{1}{2} g_{AB}\dot{q}^A\dot{q}^B$. The metric inherits block structure from (3.22): a rotational block governed by the Watson reciprocal inertia $\boldsymbol{\mu}(Q) = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$, an identity vibrational block (normal coordinates), and a Cartesian block for $\bar{\mathbf{r}}_i$.

4.1.1 Podolsky form / Laplace–Beltrami quantization

The natural Hilbert space is $L^2(\mathcal{C}, \sqrt{|g|}\, d^n q)$, where the Riemannian volume element $\sqrt{|g|}\, d^n q$ is the unique measure compatible with the metric. To produce a Hermitian kinetic operator on this Hilbert space, replace T by the Laplace–Beltrami operator divided by $-2/\hbar^2$:

$$\boxed{\hat{T} \;=\; -\frac{\hbar^2}{2}\, \frac{1}{\sqrt{|g|}}\, \partial_A\big(\sqrt{|g|}\, g^{AB}\, \partial_B\big).} \tag{4.3}$$

This is the *Podolsky prescription* [13]. Equivalently, one may write $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ and then take $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB} \hat{\pi}_B$ in symmetric (Weyl-type) ordering; the two prescriptions coincide up to terms absorbed into the volume element.

For our problem, only the Euler-angle and Q_b sectors are curvilinear; the BF Cartesian electronic sector retains the standard $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}}\bar{\mathbf{r}})$.

4.2 Volume element and Podolsky rescaling

4.2.1 Configuration-space metric and the volume element

The configuration-space metric requires the kinetic energy with *all* coordinates — the electrons included — carried as velocities $\mathbf{v} = (\boldsymbol{\omega}, \dot{\mathbf{Q}}, \dot{\bar{\mathbf{r}}}_i)$. (This velocity-form representation, used here solely to read off the measure, is distinct from the Routhian (3.13), in which the electrons are kept in momentum form.) Expanding the body-fixed nuclear, electronic, and mass-polarization kinetic terms via (3.9d)–(3.9e), in normal coordinates (vibrational block $\sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$),

$$2T = \mathbf{v}^{\mathsf{T}}\mathsf{M}\,\mathbf{v}, \qquad \mathsf{M} = \begin{pmatrix} \mathbf{I} & \mathsf{Z} & \mathsf{C} \\ \mathsf{Z}^{\mathsf{T}} & \mathbb{1} & 0 \\ \mathsf{C}^{\mathsf{T}} & 0 & \mathsf{D} \end{pmatrix}, \tag{4.3a}$$

with $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ the full instantaneous inertia ($\boldsymbol{\omega}$–$\boldsymbol{\omega}$ block), $\mathsf{Z}_{\alpha b} = (\boldsymbol{\sigma}_b)_\alpha$ the rotation–vibration Coriolis block, C the rotation–electron block (from $\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X)$), and $\mathsf{D}_{(i\alpha)(j\beta)} = (m_e\delta_{ij} - m_e^2/M_{\rm tot})\,\delta_{\alpha\beta}$ the *constant* electron block. The physical velocities relate to the coordinate velocities $\dot q = (\dot\phi, \dot\theta, \dot\chi, \dot Q_b, \dot{\bar{\mathbf{r}}}_i)$ by $\mathbf{v} = \mathsf{A}(q)\dot q$ with $\mathsf{A} = \mathrm{diag}\big(\mathsf{E}(\Omega), \mathbb{1}, \mathbb{1}\big)$, where $\boldsymbol{\omega} = \mathsf{E}(\Omega)\dot{\boldsymbol{\Omega}}$ is the Euler kinematic relation. Hence $g_{AB} = (\mathsf{A}^{\mathsf{T}}\mathsf{M}\mathsf{A})_{AB}$ and

$$\det g = (\det \mathsf{A})^2 \det \mathsf{M} = \sin^2\theta\ \det \mathsf{M}, \qquad |\det \mathsf{E}(\Omega)| = \sin\theta. \tag{4.3b}$$

Reduce $\det \mathsf{M}$ by the Schur complement of the constant electron block D:

$$\det \mathsf{M} = \det \mathsf{D}\ \det\begin{pmatrix} \mathbf{I} - \mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} & \mathsf{Z} \\ \mathsf{Z}^{\mathsf{T}} & \mathbb{1} \end{pmatrix}. \tag{4.3c}$$

The electron Schur correction is precisely the electronic + mass-polarization inertia $\mathbf{I}_e - \mathbf{I}_X$ of (3.9e): completing the square in $\dot{\bar{\mathbf{r}}}_i$ (equivalently, eliminating the electron velocities in favour of the momenta $\bar{\mathbf{p}}_i$) removes all $\boldsymbol{\omega}$-dependence from the electron block, $\mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_e - \mathbf{I}_X$, so $\mathbf{I} - \mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_N$ — the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the cross-check (3.18). A second Schur complement, now of the $\mathbb{1}$ vibrational block, gives

$$\det \mathsf{M} = \det \mathsf{D}\ \det\big(\mathbf{I}_N - \mathsf{Z}\mathsf{Z}^{\mathsf{T}}\big) = \det \mathsf{D}\ \det \mathbf{I}', \qquad \mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}, \tag{4.3d}$$

using $\mathsf{Z}\mathsf{Z}^{\mathsf{T}} = \sum_b \boldsymbol{\sigma}_b\boldsymbol{\sigma}_b^{\mathsf{T}} = \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$, with $\det \mathsf{D}$ a constant. The Coriolis coupling therefore enters the determinant through the *same* modified inertia $\mathbf{I}'$ that defines the kernel $\boldsymbol{\mu} = (\mathbf{I}')^{-1}$. Dropping the constant electron factor (an overall normalization of the flat electronic measure), the Riemannian volume element is

$$\boxed{\sqrt{|g|}\,d^n q \;=\; \underbrace{\sin\theta\,d\phi\,d\theta\,d\chi}_{\text{Haar on } SO(3)} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)}\,\prod_b dQ_b}_{\text{rot.-vib. Jacobian}} \cdot \underbrace{\prod_i d^3\bar{\mathbf{r}}_i}_{\text{flat (const.)}}\ .} \tag{4.4}$$

The Jacobian carries the Watson modified inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$, *not* the bare $\mathbf{I}_N$: the rotation–vibration Coriolis coupling reduces the rotational determinant from $\det \mathbf{I}_N$ to $\det(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})$.

4.2.2 Wavefunction rescaling

To work with the simpler measure $d\mu_W \equiv \sin\theta\,d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$, define the rescaled wavefunction

$$\psi_W \;\equiv\; \big(\det \mathbf{I}'(Q)\big)^{1/4}\,\psi. \tag{4.5}$$

Then $|\psi|^2\sqrt{|g|}\,d^n q = |\psi_W|^2\,d\mu_W$, so $\psi_W \in L^2(\mathcal{C}, d\mu_W)$. The kinetic operator transforms by a similarity: $\hat T_W = \mathcal{J}^{1/2}\hat T\mathcal{J}^{-1/2}$ with $\mathcal{J} = (\det \mathbf{I}')^{1/2}$. The residual non-cancellation generates a new Q-dependent ordinary potential — the Watson pseudopotential (next subsection).

4.3 Anomalous BF angular-momentum algebra

Total angular momentum is naturally defined as an SF observable; its BF components are obtained by projection through $\mathbf{S}$:

$$\hat{J}^{\mathrm{BF}}_\alpha \;=\; \mathbf{S}_{\beta\alpha}\,\hat{J}^{\mathrm{SF}}_\beta. \tag{4.6}$$

The space-fixed components are ordinary angular-momentum operators in the Euler angles; they obey the normal algebra and act on the direction-cosine matrix $\mathbf{S} = \mathbf{S}(\Omega)$ through its *space-fixed* index,

$$[\hat{J}^{\mathrm{SF}}_\gamma, \hat{J}^{\mathrm{SF}}_\delta] = i\hbar\,\epsilon_{\gamma\delta\lambda}\hat{J}^{\mathrm{SF}}_\lambda, \qquad [\hat{J}^{\mathrm{SF}}_\gamma, S_{\delta\beta}] = i\hbar\,\epsilon_{\gamma\delta\lambda}S_{\lambda\beta}. \tag{4.6a}$$

Because $\mathbf{S}$ is a proper rotation ($\det\mathbf{S} = 1$), the triple product of direction cosines collapses to one Levi-Civita symbol, and orthogonality inverts the projection (4.6):

$$\epsilon_{\gamma\delta\lambda}\,S_{\gamma\alpha}S_{\delta\beta}S_{\lambda\nu} = \det(\mathbf{S})\,\epsilon_{\alpha\beta\nu} = \epsilon_{\alpha\beta\nu}, \qquad \hat{J}^{\mathrm{SF}}_\lambda = S_{\lambda\nu}\,\hat{J}^{\mathrm{BF}}_\nu. \tag{4.6b}$$

Now evaluate the body-fixed commutator (writing $\hat{J} \equiv \hat{J}^{\mathrm{SF}}$), moving the Euler-angle functions $S_{\delta\beta}$ leftward past $\hat{J}_\gamma$ with (4.6a):

$$\begin{aligned}
[\hat{J}^{\mathrm{BF}}_\alpha, \hat{J}^{\mathrm{BF}}_\beta] &= [S_{\gamma\alpha}\hat{J}_\gamma,\, S_{\delta\beta}\hat{J}_\delta] = S_{\gamma\alpha}S_{\delta\beta}[\hat{J}_\gamma, \hat{J}_\delta] + S_{\gamma\alpha}\,[\hat{J}_\gamma, S_{\delta\beta}]\,\hat{J}_\delta - S_{\delta\beta}\,[\hat{J}_\delta, S_{\gamma\alpha}]\,\hat{J}_\gamma \\
&= i\hbar\,\epsilon_{\gamma\delta\lambda}\big(S_{\gamma\alpha}S_{\delta\beta}\hat{J}_\lambda + S_{\gamma\alpha}S_{\lambda\beta}\hat{J}_\delta + S_{\delta\beta}S_{\lambda\alpha}\hat{J}_\gamma\big),
\end{aligned}$$

where $[\hat{J}_\gamma, \hat{J}_\delta] = i\hbar\epsilon_{\gamma\delta\lambda}\hat{J}_\lambda$, $[\hat{J}_\gamma, S_{\delta\beta}] = i\hbar\epsilon_{\gamma\delta\lambda}S_{\lambda\beta}$, and $-[\hat{J}_\delta, S_{\gamma\alpha}] = -i\hbar\epsilon_{\delta\gamma\lambda}S_{\lambda\alpha} = +i\hbar\epsilon_{\gamma\delta\lambda}S_{\lambda\alpha}$ were collected under a common $\epsilon_{\gamma\delta\lambda}$. Insert $\hat{J} = \mathbf{S}\hat{J}^{\mathrm{BF}}$ in each term and apply (4.6b):

$$\epsilon_{\gamma\delta\lambda}S_{\gamma\alpha}S_{\delta\beta}\,\hat{J}_\lambda = \epsilon_{\alpha\beta\nu}\hat{J}^{\mathrm{BF}}_\nu, \quad \epsilon_{\gamma\delta\lambda}S_{\gamma\alpha}S_{\lambda\beta}\,\hat{J}_\delta = \epsilon_{\alpha\nu\beta}\hat{J}^{\mathrm{BF}}_\nu, \quad \epsilon_{\gamma\delta\lambda}S_{\delta\beta}S_{\lambda\alpha}\,\hat{J}_\gamma = \epsilon_{\nu\beta\alpha}\hat{J}^{\mathrm{BF}}_\nu.$$

Since $\epsilon_{\alpha\beta\nu} + \epsilon_{\alpha\nu\beta} + \epsilon_{\nu\beta\alpha} = \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} = -\epsilon_{\alpha\beta\nu}$, the three pieces sum to a single term of *reversed* sign — the extra contribution from differentiating $\mathbf{S}$ outweighs and flips the bare SF algebra:

$$\boxed{[\hat{J}^{\mathrm{BF}}_\alpha, \hat{J}^{\mathrm{BF}}_\beta] \;=\; -i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{J}^{\mathrm{BF}}_\gamma.} \tag{4.7}$$

This is the famous **anomalous commutator** for BF angular momentum [14, 7], reviewed in Bunker–Jensen [4].

In contrast, BF electronic angular momentum is built intrinsically from BF Cartesian electron operators:

$$[\hat{L}_{\mathrm{elec},\alpha}, \hat{L}_{\mathrm{elec},\beta}] \;=\; +i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{L}_{\mathrm{elec},\gamma}, \tag{4.8}$$

the *normal* algebra. Since $\hat{J}^{\mathrm{BF}}$ acts only on Euler angles and $\hat{\mathbf{L}}_{\mathrm{elec}}$ acts only on BF electron coordinates, the two commute:

$$[\hat{J}^{\mathrm{BF}}_\alpha, \hat{L}_{\mathrm{elec},\beta}] \;=\; 0. \tag{4.9}$$

The vibrational $\hat{\pi}_\alpha$, built from Q_b and $\hat{P}_b$, also commutes with both $\hat{J}^{\mathrm{BF}}$ and $\hat{\mathbf{L}}_{\mathrm{elec}}$. From here on we write $\hat{J}_\alpha \equiv \hat{J}_\alpha^{\mathrm{BF}}$ and the anomalous algebra (4.7) is understood.

4.4 Watson pseudopotential

We now quantize the rovibrational kinetic energy of (3.22),

$$T_{\mathrm{rovib}} = \tfrac{1}{2}\sum_{\alpha\beta} A_\alpha\,\mu_{\alpha\beta}(Q)\,A_\beta + \tfrac{1}{2}\sum_b P_b^2, \qquad A_\alpha \equiv J_\alpha - L_{\mathrm{elec},\alpha} - \pi_\alpha, \tag{4.10a}$$

a quadratic form in the momenta (J_α, P_b) whose coefficients $\mu_{\alpha\beta}(Q)$ depend on the shape. The naïve replacement $P_b \to -i\hbar\,\partial_{Q_b}$ is *not* Hermitian on the curved measure (4.4); the Podolsky (Laplace–Beltrami) prescription (4.3) restores Hermiticity, and the rescaling (4.5) to the flat measure then leaves a c-number residue — the **Watson pseudopotential** $\hat{U}(Q)$. We carry out the reduction in full.

4.4.1 The curvilinear extra-potential, in general

For arbitrary coordinates q^A with covariant metric g_{AB} ($2T = g_{AB}\dot{q}^A\dot{q}^B$), contravariant inverse g^{AB}, and determinant $g \equiv \det g_{AB}$, the kinetic operator Hermitian on $L^2(\sqrt{g}\,d^nq)$ is the Laplace–Beltrami operator

$$\hat{T}_{\mathrm{LB}} = -\frac{\hbar^2}{2}\,g^{-1/2}\,\partial_A\big(g^{1/2}\,g^{AB}\,\partial_B\big). \tag{4.10b}$$

To pass to the flat measure $\prod_A dq^A$, set $\psi = g^{-1/4}\psi_W$ (so $\psi_W = g^{1/4}\psi$) and conjugate, $\hat{T}_W = g^{1/4}\hat{T}_{\mathrm{LB}}\,g^{-1/4}$. Writing $\rho \equiv g^{1/2}$ one has $\partial_B\psi = \rho^{-1/2} f_B$ with $f_B \equiv \partial_B\psi_W - \frac{1}{2}(\partial_B \ln\rho)\psi_W$; carrying the $g^{1/4}$ through $\hat{T}_{\mathrm{LB}}$,

$$\hat{T}_W\psi_W = -\frac{\hbar^2}{2}\Big[\underbrace{\tfrac{1}{2}(\partial_A \ln\rho)\,g^{AB} f_B}_{\text{(i)}} + \underbrace{(\partial_A g^{AB})\,f_B}_{\text{(ii)}} + \underbrace{g^{AB}\,\partial_A f_B}_{\text{(iii)}}\Big]. \tag{4.10c}$$

Insert f_B and $\partial_A f_B = \partial_A\partial_B\psi_W - \frac{1}{2}(\partial_A\partial_B\ln\rho)\psi_W - \frac{1}{2}(\partial_B\ln\rho)\partial_A\psi_W$, and sort by the order of derivative acting on ψ_W:

$\partial\partial\psi_W$: only (iii) contributes, $g^{AB}\partial_A\partial_B\psi_W$;

$\partial\psi_W$: (i) gives $+\frac{1}{2}g^{AB}(\partial_A\ln\rho)\partial_B\psi_W$ and (iii) gives $-\frac{1}{2}g^{AB}(\partial_B\ln\rho)\partial_A\psi_W$, which *cancel* by symmetry of g^{AB}, leaving only $(\partial_A g^{AB})\partial_B\psi_W$ from (ii);

ψ_W : (i) gives $-\frac{1}{4}g^{AB}(\partial_A\ln\rho)(\partial_B\ln\rho)$, (ii) gives $-\frac{1}{2}(\partial_A g^{AB})(\partial_B\ln\rho)$, (iii) gives $-\frac{1}{2}g^{AB}\partial_A\partial_B\ln\rho$.

The first two derivative-orders reassemble into $\partial_A(g^{AB}\partial_B\psi_W)$, i.e. the flat symmetric operator $\frac{1}{2}\hat{p}_A g^{AB}\hat{p}_B$ ($\hat{p}_A = -i\hbar\partial_A$). The ψ_W-terms, using $-\frac{1}{2}(\partial_A g^{AB})(\partial_B\ln\rho) - \frac{1}{2}g^{AB}\partial_A\partial_B\ln\rho = -\frac{1}{2}\partial_A(g^{AB}\partial_B\ln\rho)$ and $\ln\rho = \frac{1}{2}\ln g$, give the extra-potential:

$$\boxed{\hat{T}_W = \tfrac{1}{2}\,\hat{p}_A\,g^{AB}\,\hat{p}_B + U, \qquad U = \frac{\hbar^2}{8}\Big[\tfrac{1}{4}\,g^{AB}(\partial_A\ln g)(\partial_B\ln g) + \partial_A\big(g^{AB}\partial_B\ln g\big)\Big].} \tag{4.10d}$$

Equation (4.10d) is exact and coordinate-general; the prefactor $\hbar^2/8$ traces to the two half-powers $g^{\pm 1/4}$ of the rescaling.

4.4.2 The molecular metric and its determinant

Read the contravariant metric off (4.10a). With $\pi_\alpha = \sum_b \sigma_{b\alpha} P_b$ (i.e. $\boldsymbol{\pi} = \boldsymbol{\sigma}^\mathsf{T} P$, $\sigma_{b\alpha} = (\boldsymbol{\sigma}_b)_\alpha$), and dropping the electronic shift $\hat{\mathbf{L}}_{\text{elec}}$ (a different sector, commuting with the nuclear metric),

$$2T_{\text{rovib}} = J^\mathsf{T}\boldsymbol{\mu}J - 2J^\mathsf{T}\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T}P + P^\mathsf{T}(\mathbb{1} + \boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T})P,$$

so in the body-frame momentum basis (J_α, P_b),

$$g^{AB} = \begin{pmatrix} \boldsymbol{\mu} & -\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T} \\ -\boldsymbol{\sigma}\boldsymbol{\mu} & \mathbb{1} + \boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T} \end{pmatrix}. \tag{4.10e}$$

By the Schur complement of the $\boldsymbol{\mu}$ block,

$$\det g^{AB} = \det\boldsymbol{\mu}\ \det\!\big(\mathbb{1} + \boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T} - \boldsymbol{\sigma}\boldsymbol{\mu}\,\boldsymbol{\mu}^{-1}\boldsymbol{\mu}\boldsymbol{\sigma}^\mathsf{T}\big) = \det\boldsymbol{\mu}, \tag{4.10f}$$

hence $g = \det g_{AB} = 1/\det\boldsymbol{\mu} = \det\mathbf{I}' \equiv \gamma(Q)$, consistent with $\sqrt{|g|} = \sin\theta\sqrt{\gamma}$ of (4.4). The full measure carries $\ln g = 2\ln\sin\theta + \ln\gamma$. In the rovibrational convention the body-frame $\hat{J}_\alpha$ are already Hermitian on $\sin\theta\, d\Omega$ [with the anomalous algebra (4.7)], so the $2\ln\sin\theta$ is accounted for by the $\hat{J}_\alpha$; *only* $\gamma(Q)$ is removed by the rescaling (4.5), and we set $\ln g \to \ln\gamma$ (a function of Q alone) in (4.10d).

4.4.3 Rotational sector: no c-number

The only operator ambiguity in the rotational kernel is the ordering of $\hat{J}_\alpha\hat{J}_\beta$. Splitting into symmetric and antisymmetric parts and using the anomalous algebra (4.7),

$$\tfrac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta = \tfrac{1}{4}\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} + \tfrac{1}{4}\mu_{\alpha\beta}[\hat{J}_\alpha, \hat{J}_\beta] = \tfrac{1}{4}\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} - \frac{i\hbar}{4}\,\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma. \tag{4.10g}$$

Since $\mu_{\alpha\beta} = \mu_{\beta\alpha}$ is symmetric while $\epsilon_{\alpha\beta\gamma}$ is antisymmetric in (α,β), the contraction $\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$: the commutator term vanishes, the asymmetric-top operator is Hermitian as written, and it contributes *no* c-number. The same holds for the full kernel $\frac{1}{2}\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta$ ($\hat{A}_\alpha = \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha$), since $\boldsymbol{\mu}$ is symmetric and commutes with $\hat{J}$ and $\hat{\mathbf{L}}_{\text{elec}}$.

4.4.4 Vibrational sector: the rescaling, step by step

With $g^{ab} = \delta_{ab}$ but the measure carrying $\gamma(Q)$, the vibrational part of (4.10b) is

$$\hat{T}_{\text{vib}} = -\frac{\hbar^2}{2}\,\gamma^{-1/2}\sum_b \partial_{Q_b}\big(\gamma^{1/2}\,\partial_{Q_b}\big). \tag{4.10h}$$

Conjugate by $\gamma^{1/4}$ and act on ψ_W, working from the innermost derivative outward (each line is one application of the product rule):

$$\begin{aligned}
\partial_{Q_b}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{-1/4}\Big[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W\Big],\\
\gamma^{1/2}\,\partial_{Q_b}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{1/4}\Big[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W\Big],\\
\partial_{Q_b}\Big(\gamma^{1/4}\big[\cdots\big]\Big) &= \gamma^{1/4}\Big[\partial^2_{Q_b}\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\psi_W - \tfrac{1}{4}(\partial^2_{Q_b}\ln\gamma)\psi_W\Big],
\end{aligned} \tag{4.10i}$$

where in the last line the two cross terms $+\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}\psi_W$ (from differentiating $\gamma^{1/4}$) and $-\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}\psi_W$ (from the bracket) cancel. Multiplying by $-\frac{\hbar^2}{2}\gamma^{1/4}\cdot\gamma^{-1/2} = -\frac{\hbar^2}{2}\gamma^{-1/4}$ and summing over b,

$$\gamma^{1/4}\hat{T}_{\rm vib}\gamma^{-1/4} = \tfrac{1}{2}\sum_b \hat{P}_b^2 + U_{\rm vib}, \qquad \hat{P}_b = -i\hbar\,\partial_{Q_b}, \tag{4.10j}$$

with the *Pauli residue*

$$U_{\rm vib} = \frac{\hbar^2}{8}\sum_b\Big[\partial_{Q_b}^2\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\Big] = \frac{\hbar^2}{2}\,\gamma^{-1/4}\Big(\sum_b\partial_{Q_b}^2\Big)\gamma^{1/4}. \tag{4.10k}$$

This is exactly what (4.10d) yields for the *flat* vibrational block $g^{ab} = \delta_{ab}$; the off-diagonal $-\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}}$ and the $\boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}}$ dressing of (4.10e) supply the remaining *Coriolis* c-numbers. Collecting the three sectors,

$$\begin{aligned}\hat{T}_{\rm rovib} \;&= \tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{{\rm elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{{\rm elec},\beta} - \hat{\pi}_\beta\big)\\ &\quad + \tfrac{1}{2}\sum_b\hat{P}_b^2 \;+\; \hat{U}(Q), \qquad \hat{U} = U_{\rm vib} + U_{\rm Cor}.\end{aligned} \tag{4.10}$$

4.4.5 Collapse to the Watson trace

The residue (4.10k) and the Coriolis c-numbers $U_{\rm Cor}$ collapse by two exact identities. First, Jacobi's formula for $\gamma = \det\mathbf{I}'$ and the derivative of an inverse,

$$\partial_b\ln\gamma = {\rm tr}\big(\boldsymbol{\mu}\,\partial_b\mathbf{I}'\big) = \sum_{\alpha\beta}\mu_{\alpha\beta}\,\partial_b\mathbf{I}'_{\alpha\beta}, \qquad \partial_b\boldsymbol{\mu} = -\,\boldsymbol{\mu}\,(\partial_b\mathbf{I}')\,\boldsymbol{\mu}, \tag{4.10l}$$

re-express every $\partial_b\ln\gamma$ and $\partial_b\boldsymbol{\mu}$ through $\partial_b\mathbf{I}'$. Second, since $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$ with $\boldsymbol{\sigma}_b = \sum_a\boldsymbol{\zeta}_{ab}Q_a$ *linear* in Q [(3.17a)],

$$\partial_c\mathbf{I}'_{\alpha\beta} = -\sum_b\big(\zeta^\alpha_{cb}\,\sigma_{b\beta} + \sigma_{b\alpha}\,\zeta^\beta_{cb}\big), \qquad \zeta^\alpha_{cb} \equiv (\boldsymbol{\zeta}_{cb})_\alpha = \partial_c\sigma_{b\alpha}\ \text{(constant)}, \tag{4.10m}$$

so $\partial_b\mathbf{I}'$ is built from the *constant* Coriolis matrices $\boldsymbol{\zeta}_{cb} = \sum_j\mathbf{l}_{jc}\times\mathbf{l}_{jb}$. Substituting (4.10l)–(4.10m) into the vibrational residue (4.10k) and into $U_{\rm Cor}$, the gradient terms $\propto\partial_b\,{\rm tr}\,\boldsymbol{\mu}$ generated by the two groups cancel identically; the surviving Q-local remainder is fixed by the Coriolis sum rules obeyed by the $\boldsymbol{\zeta}$'s [from the orthonormality $\sum_j\mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$ of the Eckart mode vectors, the $\boldsymbol{\zeta}^{(\alpha)}$ acting as $SO(3)$ generators on the mode index], which collapse it to a single trace over the three Cartesian components [3]:

$$\boxed{\hat{U}(Q) \;=\; -\frac{\hbar^2}{8}\sum_{\alpha=1}^{3}\mu_{\alpha\alpha}(Q), \qquad \boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}.} \tag{4.11}$$

Two remarks. *(a)* The cancellation is special to *rectilinear* normal coordinates: for general curvilinear Q the linear relation (4.10m) fails, the residue (4.10k) does *not* collapse to the trace, and additional vibrational extrapotential terms survive. *(b)* The electrons do not modify $\hat{U}$: $\mu_{\alpha\beta}$ depends on Q alone, the electronic sector enters the rovib block only through $\hat{\mathbf{L}}_{\text{elec}}$ (which commutes with $\boldsymbol{\mu}$ and with the Q-measure γ), and the electronic Cartesian Jacobian is flat. The pseudopotential is a c-number on the nuclear configuration space, fixed entirely by the modified inertia $\mathbf{I}'$.

4.4.6 Algebraic cross-check of the coefficient

The $-\hbar^2/8$ is confirmed by the equivalent algebraic (Weyl-ordering) route noted under (4.3): with the Hermitian momenta $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln\sqrt{|g|})$ one has $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB}\hat{\pi}_B$, identical to the Laplace–Beltrami operator. Three observations fix the result:

- *Vibrational sector.* With $g^{ab} = \delta_{ab}$ and $\sqrt{|g|} \propto \gamma^{1/2}$, expanding $\frac{1}{2}\sum_b \hat{\pi}_{Q_b}^2$ (the Hermitian momenta of (4.3) specialized to the modes) and rescaling reproduces exactly the residue (4.10k), $U_{\text{vib}} = \frac{\hbar^2}{2}\gamma^{-1/4}\big(\sum_b \partial_{Q_b}^2\big)\gamma^{1/4}$ — the $-\hbar^2/8$ scale enters through the two $\frac{1}{2}\partial_A \ln\sqrt{|g|}$ factors, $\frac{1}{2}\hat{\pi}_{Q_b}^2 \supset \frac{\hbar^2}{8}(\partial_{Q_b}\ln\gamma)^2$ etc.
- *Rotational sector.* $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$, so the only reordering is among the $\hat{J}$'s; by the anomalous algebra (4.7) the antisymmetric part of $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$ vanishes ($\mu_{\alpha\beta} = \mu_{\beta\alpha}$), so this block adds no independent c-number.
- *Reduction.* Carrying the Watson cancellations (Jacobi's formula and the Coriolis sum rules) through both sectors collapses every Q-derivative to the single trace $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$, in agreement with (4.11).

The pseudopotential is a c-number depending only on Q (a function on the nuclear configuration space).

4.4.7 Operator ordering: what is and isn't ambiguous

- $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ (different coordinate sectors: Euler angles vs. Q), with $\hat{\mathbf{L}}_{\text{elec}}$ (BF electron coords), but not with $\hat{P}_b$ (since $\partial\mu_{\alpha\beta}/\partial Q_b \neq 0$). Hence $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta = \mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$.
- Within the rotational block, the symmetric product $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$ in (4.10) is unambiguous because μ is symmetric in (α, β), and the Watson pseudopotential (4.11) accounts for the residual non-commutativity of the $\hat{J}$'s themselves.
- In normal coordinates the vibrational term is the bare $\frac{1}{2}\sum_b \hat{P}_b^2$; the Q-dependence of the volume element has been moved into $\hat{U}(Q)$ by the rescaling (4.5), which is precisely what Podolsky produces.

4.5 Electronic kinetic operator

In BF Cartesian electron coordinates, $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}}\bar{\mathbf{r}})$. The electronic + mass-polarization kinetic operator is therefore the straightforward quantization of (3.21):

$$\hat{T}_e = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2. \tag{4.12}$$

Expanding the second piece:

$$\frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2M_N} + \frac{1}{M_N}\sum_{i<l} \hat{\bar{\mathbf{p}}}_i \cdot \hat{\bar{\mathbf{p}}}_l. \tag{4.13}$$

The diagonal $1/(2M_N)$ piece renormalizes the effective electron mass through $1/m_e + 1/M_N = M_{\rm tot}/(m_e M_N) = 1/\mu_e$. The off-diagonal two-electron term is the mass-polarization *sensu stricto*, dynamically coupling electrons through nuclear recoil — a purely classical effect (Section 2.4) inherited by the quantum theory.

4.6 Complete quantum molecular Hamiltonian

Combining (4.10), (4.11), (4.12), and the rotationally invariant Coulomb potentials from (3.8):

$$\boxed{\begin{aligned} \hat H_{\rm int} \;&= \tfrac12 \sum_{\alpha\beta} \big(\hat J_\alpha - \hat L_{{\rm elec},\alpha} - \hat\pi_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat J_\beta - \hat L_{{\rm elec},\beta} - \hat\pi_\beta\big) \\ &\quad + \tfrac12 \sum_b \hat P_b^2 \\ &\quad + \sum_i \frac{\hat{\bar{\mathbf p}}_i^{\,2}}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf p}}_i\Big)^2 \\ &\quad - \frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q) \\ &\quad + V_{NN}(Q) \;+\; V_{Ne}(Q,\{\bar{\mathbf r}_i\}) \;+\; V_{ee}(\{\bar{\mathbf r}_i\}), \end{aligned}} \tag{4.14}$$

acting on $L^2(\mathcal C,\, d\mu_W)$ with $d\mu_W = \sin\theta\, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf r}_i$. All operators are Hermitian on this Hilbert space.

This is the **exact non-relativistic molecular Hamiltonian** after separation of the total COM motion, in BF + Eckart frame, with electrons referred to the nuclear COM. The only approximations made along the way were: (i) non-relativistic dynamics; (ii) Eckart's choice of the BF orientation (a convention, not an approximation); (iii) parametrization of the nuclear shape by $3N_{\rm nucl} - 6$ internal coordinates Q_b (a parametrization, also exact).

4.7 Born–Huang expansion

4.7.1 Clamped-nuclei electronic Hamiltonian

At each nuclear configuration Q, define the **BF clamped-nuclei electronic Hamiltonian** as the sum of all Q-parametric electron-only pieces of (4.14):

$$\hat H_{\rm el}(Q) \;\equiv\; \sum_i \frac{\hat{\bar{\mathbf p}}_i^{\,2}}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf p}}_i\Big)^2 \;+\; V_{Ne}(Q,\{\bar{\mathbf r}_i\}) \;+\; V_{ee}(\{\bar{\mathbf r}_i\}). \tag{4.15}$$

Remark. The mass-polarization piece $(\sum_i \hat{\bar{\mathbf p}}_i)^2/(2M_N)$ is included here as a small correction with $1/M_N$. Some treatments ("clamped-nuclei BO") drop it; we keep it explicit because it is parametric in Q only through the absent dependence (i.e., M_N is just a number), which means it acts trivially on the parametric family of electronic Hilbert spaces.

Solve the parametric eigenvalue problem:

$$\hat H_{\rm el}(Q)\,|\phi_n(Q)\rangle \;=\; E_n^{\rm el}(Q)\,|\phi_n(Q)\rangle, \tag{4.16}$$

in the electronic Hilbert space $\mathcal H_e = L^2(\mathbb R^{3N_{\rm elec}}, d^{3N_{\rm elec}}\bar{\mathbf r})$. The eigenstates $\{|\phi_n(Q)\rangle\}$ depend *parametrically* on Q — Q is treated as a fixed classical parameter, not a quantum degree of freedom — and form a complete orthonormal basis of $\mathcal{H}_e$ at each Q:

$$\sum_n |\phi_n(Q)\rangle\langle\phi_n(Q)| \;=\; \hat{\mathbb{1}}_e, \qquad \langle\phi_m(Q)|\phi_n(Q)\rangle_e = \delta_{mn}. \tag{4.17}$$

4.7.2 The exact Born–Huang expansion

The *exact* molecular wavefunction $|\Psi\rangle \in L^2(\mathcal{C}, d\mu_W)$ admits a resolution against the parametric electronic basis. Insert the resolution of identity (4.17) at each Q:

$$|\Psi\rangle \;=\; \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle, \qquad \chi_n(\Omega, Q) \;\equiv\; \langle\phi_n(Q)|\Psi\rangle_e, \tag{4.18}$$

where $\Omega = (\phi, \theta, \chi)$, the electronic inner product is taken at fixed Q, and χ_n are the **nuclear coefficient functions**. The expansion (4.18) is **exact**, not an ansatz: it follows from the completeness (4.17) at each Q, with the only subtlety being that the basis $|\phi_n(Q)\rangle$ varies parametrically. There is *no* simple tensor factorization $\mathcal{H}_{\text{molecule}} = \mathcal{H}_N \otimes \mathcal{H}_e$ because of this parametric dependence; the Born–Huang expansion [11] is the rigorous replacement.

4.7.3 Coupled-channel equations

Substitute (4.18) into $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$, project from the left by $\langle\phi_m(Q)|$, and use (4.16):

$$\big[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n(\Omega, Q) \;+\; \sum_m \hat{\Lambda}_{nm}\,\chi_m(\Omega, Q) \;=\; E\,\chi_n(\Omega, Q), \tag{4.19}$$

where $\hat{T}_N$ denotes the nuclear (rotational + vibrational) kinetic operator from the first two lines of (4.14). The **non-adiabatic coupling operators** $\hat{\Lambda}_{nm}$ arise from the action of $\hat{T}_N$ (and the rot–electronic couplings through $\hat{\mathbf{L}}_{\text{elec}}$) on the Q-parametric dependence of $|\phi_m(Q)\rangle$:

$$\hat{\Lambda}_{nm} \;\sim\; -\frac{\hbar^2}{1}\,\langle\phi_n|\nabla_Q\phi_m\rangle_e\cdot\nabla_Q \;-\; \frac{\hbar^2}{2}\,\langle\phi_n|\nabla_Q^2\phi_m\rangle_e \;+\; \dots, \tag{4.20}$$

with analogous terms from the rotation–electron Coriolis $\hat{J}_\alpha \hat{L}_{\text{elec},\alpha}$ kernel.

4.7.4 Born–Oppenheimer approximation as truncation

The **Born–Oppenheimer (BO) approximation** is the diagonal truncation of (4.19): set $\hat{\Lambda}_{nm} \to 0$ for $m \neq n$ and (optionally) also $\hat{\Lambda}_{nn} \to 0$. The resulting effective nuclear problem is

$$\big[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n^{\text{BO}}(\Omega, Q) \;=\; E^{\text{BO}}\,\chi_n^{\text{BO}}(\Omega, Q), \tag{4.21}$$

where the **Born–Oppenheimer potential energy surface** for electronic state n is

$$V_n^{\text{BO}}(Q) \;\equiv\; E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \tag{4.22}$$

The exactness of (4.18) and the well-definedness of (4.19) are mathematical facts. BO is the physical assumption that off-diagonal $\hat{\Lambda}_{nm}$ are small — justified when electronic gaps are large compared to nuclear kinetic energies, breaking down at conical intersections and avoided crossings, where the parametric variation of $|\phi_n(Q)\rangle$ becomes singular.

4.7.5 Summary

1. (4.14) is the exact non-relativistic quantum molecular Hamiltonian in BF + Eckart + nuclear-COM coordinates.
2. The Born–Huang expansion (4.18) is exact, following from completeness of the parametric electronic basis (4.17).
3. The infinite coupled system (4.19) is equivalent to the Schrödinger equation $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$.
4. The Born–Oppenheimer truncation reduces it to (4.21), giving the familiar single-PES picture (4.22).

Appendix

Appendix A: Mass-Polarization Term in Classical Molecular Hamiltonian

We start from the laboratory-frame kinetic energy

$$T = \frac{1}{2}\sum_j M_j \dot{\mathbf{R}}_j^2 + \frac{1}{2}\sum_i m_e \dot{\mathbf{r}}_i^2, \tag{1}$$

where M_j is the mass of the j-th nucleus, $\mathbf{R}_j$ is its Cartesian coordinate, m_e is the electron mass, and $\mathbf{r}_i$ is the coordinate of the i-th electron.

4.7.6 Nuclear Center-of-Mass Coordinate

Define the nuclear total mass

$$M_N = \sum_j M_j. \tag{2}$$

The nuclear center-of-mass coordinate is

$$\mathbf{R}_{\text{cm}} = \mathbf{R}_N^{\text{CoM}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j. \tag{3}$$

We introduce coordinates shifted by the nuclear center of mass:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \tag{4}$$

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \tag{5}$$

Therefore,

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j, \qquad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i. \tag{6}$$

The shifted nuclear coordinates satisfy the constraint

$$\sum_j M_j \mathbf{R}'_j = 0, \tag{7}$$

and therefore

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0. \tag{8}$$

This is important: the $\mathbf{R}'_j$ are not all independent.

4.7.7 Kinetic Energy in Velocity Form

Substitute

$$\dot{\mathbf{R}}_j = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j, \qquad \dot{\mathbf{r}}_i = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \tag{9}$$

into the kinetic energy. For the nuclear part,

$$T_N = \frac{1}{2}\sum_j M_j \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j\right)^2 \tag{10}$$

$$= \frac{1}{2}M_N\dot{\mathbf{R}}_{\text{cm}}^2 + \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_j M_j\dot{\mathbf{R}}'_j + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j. \tag{11}$$

Using

$$\sum_j M_j\dot{\mathbf{R}}'_j = 0, \tag{12}$$

we get

$$T_N = \frac{1}{2}M_N\dot{\mathbf{R}}_{\text{cm}}^2 + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j. \tag{13}$$

For the electronic part,

$$T_e = \frac{1}{2}\sum_i m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i\right)^2 \tag{14}$$

$$= \frac{1}{2}N_{\text{elec}}m_e\dot{\mathbf{R}}_{\text{cm}}^2 + m_e\dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2}\sum_i m_e\dot{\mathbf{r}}'^2_i, \tag{15}$$

where N_{elec} is the number of electrons. Hence the total kinetic energy is

$$T = \frac{1}{2}\left(M_N + N_{\text{elec}}m_e\right)\dot{\mathbf{R}}_{\text{cm}}^2 + m_e\dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j + \frac{1}{2}\sum_i m_e\dot{\mathbf{r}}'^2_i. \tag{16}$$

4.7.8 Canonical Momenta

The momentum conjugate to the nuclear center-of-mass coordinate $\mathbf{R}_{\text{cm}}$ is

$$\mathbf{P}_{\text{cm}} = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{R}}_{\text{cm}}} \tag{17}$$

$$= \left(M_N + N_{\text{elec}}m_e\right)\dot{\mathbf{R}}_{\text{cm}} + m_e\sum_i \dot{\mathbf{r}}'_i. \tag{18}$$

Since

$$\mathbf{p}'_i = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}'_i} = m_e\left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i\right), \tag{19}$$

we can write

$$\sum_i \mathbf{p}'_i = N_{\text{elec}}m_e\dot{\mathbf{R}}_{\text{cm}} + m_e\sum_i \dot{\mathbf{r}}'_i. \tag{20}$$

Therefore,

$$\boxed{\mathbf{P}_{\text{cm}} = M_N\dot{\mathbf{R}}_{\text{cm}} + \sum_i \mathbf{p}'_i} \tag{21}$$

or equivalently,

$$\boxed{M_N\dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i} \tag{22}$$

and hence

$$\boxed{\dot{\mathbf{R}}_{\rm cm} = \frac{1}{M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i\right).} \tag{23}$$

The electron momentum conjugate to $\mathbf{r}'_i$ is

$$\boxed{\mathbf{p}'_i = m_e\left(\dot{\mathbf{R}}_{\rm cm} + \dot{\mathbf{r}}'_i\right).} \tag{24}$$

Thus $\mathbf{p}'_i$ is the lab-frame electron momentum, not simply $m_e\dot{\mathbf{r}}'_i$. Equivalently,

$$m_e\dot{\mathbf{r}}'_i = \mathbf{p}'_i - m_e\dot{\mathbf{R}}_{\rm cm}. \tag{25}$$

The nuclear shifted momenta are

$$\boxed{\mathbf{P}'_j = M_j\dot{\mathbf{R}}'_j.} \tag{26}$$

Because the shifted nuclear coordinates satisfy

$$\sum_j M_j\mathbf{R}'_j = 0, \tag{27}$$

their conjugate momenta satisfy

$$\boxed{\sum_j \mathbf{P}'_j = 0.} \tag{28}$$

In terms of the laboratory-frame nuclear momenta

$$\mathbf{P}_j = M_j\dot{\mathbf{R}}_j, \tag{29}$$

one has

$$\boxed{\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N}\mathbf{P}_N,} \tag{30}$$

where

$$\mathbf{P}_N = \sum_j \mathbf{P}_j = M_N\dot{\mathbf{R}}_{\rm cm}. \tag{31}$$

Using

$$\mathbf{P}_N = \mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i, \tag{32}$$

we may also write

$$\boxed{\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i\right).} \tag{33}$$

4.7.9 Kinetic Energy in Momentum Form

The original kinetic energy can be written as

$$T = T_{\rm nuc,int} + T_{\rm nuc,CoM} + T_e. \tag{34}$$

The nuclear internal kinetic energy is

$$T_{\text{nuc,int}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j}. \tag{35}$$

The electron kinetic energy is

$$T_e = \sum_i \frac{\mathbf{p}_i'^2}{2m_e}. \tag{36}$$

The nuclear center-of-mass kinetic energy is

$$T_{\text{nuc,CoM}} = \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2. \tag{37}$$

Using

$$M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i', \tag{38}$$

we get

$$T_{\text{nuc,CoM}} = \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2. \tag{39}$$

Therefore the kinetic energy in terms of the new momenta is

$$\boxed{T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2.} \tag{40}$$

The accompanying constraints are

$$\boxed{\sum_j M_j \mathbf{R}_j' = 0, \qquad \sum_j \mathbf{P}_j' = 0.} \tag{41}$$

4.7.10 Expanded Form

Expanding the center-of-mass recoil term gives

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \tag{42}$$

Thus,

$$\boxed{T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2.} \tag{43}$$

4.7.11 Hamiltonian

If the potential energy is translationally invariant, then

$$V = V\left(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\} \right). \tag{44}$$

The Hamiltonian is therefore

$$\boxed{H = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i'\right)^2 + V\left(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}\right).} \tag{45}$$

4.7.12 Molecular Rest Frame

If the whole molecule is taken to be in its rest frame, then

$$\mathbf{P}_{\rm cm} = 0. \tag{46}$$

The kinetic energy becomes

$$\boxed{T_{\rm rest} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\sum_i \mathbf{p}_i'\right)^2.} \tag{47}$$

The last term,

$$\frac{1}{2M_N}\left(\sum_i \mathbf{p}_i'\right)^2, \tag{48}$$

is the nuclear center-of-mass recoil term, also called the mass-polarization term.

4.7.13 Important Interpretation

The coordinate transformation used here shifts both nuclei and electrons by the nuclear center of mass, not by the total molecular center of mass. Therefore,

$$\mathbf{p}_i' = m_e\left(\dot{\mathbf{R}}_{\rm cm} + \dot{\mathbf{r}}_i'\right) \tag{49}$$

is the canonical momentum conjugate to $\mathbf{r}_i'$. It is equal to the electron lab-frame mechanical momentum. However,

$$m_e \dot{\mathbf{r}}_i' \tag{50}$$

is only the electron momentum relative to the moving nuclear center of mass. It is not the canonical momentum conjugate to $\mathbf{r}_i'$ unless the nuclear center-of-mass motion has been fixed or eliminated. The final kinetic energy in nuclear-CoM coordinates is therefore

$$\boxed{T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i'\right)^2,} \tag{51}$$

with

$$\boxed{\sum_j M_j \mathbf{R}_j' = 0, \qquad \sum_j \mathbf{P}_j' = 0.} \tag{52}$$

Appendix B: Term-by-term quantization of the rovibrational kernel

Section 4 quantizes the classical Hamiltonian (3.22). All of the nontrivial operator-ordering work lives in one place — the **rovibrational kernel**

$$T_{\rm rovib} = \tfrac{1}{2}\sum_{\alpha\beta} A_\alpha\,\mu_{\alpha\beta}(Q)\,A_\beta, \qquad A_\alpha \equiv J_\alpha - L_{{\rm elec},\alpha} - \pi_\alpha. \tag{B.1}$$

This appendix expands the product $\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta$ into its nine pieces and quantizes each one, writing down *every* algebraic step. For each term we ask two questions: (i) what is the correct *Hermitian* ordering of the operators, and (ii) what is its explicit *differential-operator* form once $\hat{\pi}_\alpha$ is written out. Repeated Cartesian indices $\alpha, \beta, \gamma \in \{1,2,3\}$ and repeated mode indices a, b, c are summed.

B.1 Two rules and a toolkit

Rule 1 (conjugate of a product). For any two operators, $(\hat{X}\hat{Y})^\dagger = \hat{Y}^\dagger\hat{X}^\dagger$ — the order *reverses*. A real function $f(Q)$ and the basic momenta are each individually Hermitian,

$$f^\dagger = f, \qquad \hat{P}_b^\dagger = \hat{P}_b, \qquad \hat{J}_\alpha^\dagger = \hat{J}_\alpha, \qquad \hat{L}_\alpha^\dagger = \hat{L}_\alpha,$$

and an operator $\hat{O}$ is *Hermitian* (a legitimate piece of a kinetic-energy operator) when $\hat{O}^\dagger = \hat{O}$.

Rule 2 (sliding operators past each other). $\hat{X}\hat{Y} = \hat{Y}\hat{X} + [\hat{X},\hat{Y}]$. We use this to move a momentum to the other side of a Q-dependent function, picking up the commutator as a correction.

Toolkit. The three momenta live in three separate "sectors": $\hat{J}_\alpha$ acts on the Euler angles Ω, $\hat{L}_\alpha \equiv \hat{L}_{{\rm elec},\alpha}$ on the electron coordinates $\bar{\mathbf{r}}_i$, and $\hat{P}_b = -i\hbar\,\partial_{Q_b}$ on the normal coordinates Q. *Operators from different sectors commute.* Explicitly [(4.7)–(4.9)]:

$$[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\,\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma, \qquad [\hat{L}_\alpha, \hat{L}_\beta] = +i\hbar\,\epsilon_{\alpha\beta\gamma}\hat{L}_\gamma, \qquad [\hat{J}_\alpha, \hat{L}_\beta] = 0, \tag{B.2}$$

$$[\hat{J}_\alpha, f(Q)] = 0, \quad [\hat{L}_\alpha, f(Q)] = 0, \quad [\hat{J}_\alpha, \hat{P}_b] = 0, \quad [\hat{L}_\alpha, \hat{P}_b] = 0, \tag{B.3}$$

$$[\hat{P}_b, f(Q)] = -i\hbar\,\partial_{Q_b} f \implies [\hat{P}_b, \mu_{\alpha\beta}] = -i\hbar\,\partial_b\mu_{\alpha\beta}, \quad [\hat{P}_b, \sigma_{c\alpha}] = -i\hbar\,\zeta^\alpha_{bc}. \tag{B.4}$$

The last line uses the vibrational angular momentum of (3.17a)–(3.20a),

$$\hat{\pi}_\alpha = \sum_b \sigma_{b\alpha}(Q)\,\hat{P}_b, \qquad \sigma_{b\alpha} = \sum_a \zeta^\alpha_{ab}\,Q_a, \qquad \zeta^\alpha_{ab} = \Big(\textstyle\sum_j \mathbf{l}_{ja}\times\mathbf{l}_{jb}\Big)_\alpha,$$

which is *linear* in Q with *constant* coefficients ζ^α_{ab}, so $\partial_{Q_b}\sigma_{c\alpha} = \zeta^\alpha_{bc}$. Because the cross product is antisymmetric, $\zeta^\alpha_{ab} = -\zeta^\alpha_{ba}$, and therefore

$$\zeta^\alpha_{bb} = 0, \qquad \sum_b \partial_{Q_b}\sigma_{b\alpha} = \sum_b \zeta^\alpha_{bb} = 0. \tag{B.5}$$

Finally, $\mu_{\alpha\beta}(Q) = \mu_{\beta\alpha}(Q)$ is a *symmetric*, real function of Q alone.

$\hat{\pi}_\alpha$ **is Hermitian.** This one fact is used again and again below. By Rule 1, then Rule 2 with (B.4), then (B.5):

$$\hat{\pi}_\alpha^\dagger = \Big(\sum_b \sigma_{b\alpha}\hat{P}_b\Big)^\dagger = \sum_b \hat{P}_b\,\sigma_{b\alpha} = \sum_b \big(\sigma_{b\alpha}\hat{P}_b + [\hat{P}_b, \sigma_{b\alpha}]\big) = \hat{\pi}_\alpha - i\hbar \sum_b \zeta_{bb}^\alpha = \hat{\pi}_\alpha. \tag{B.6}$$

Had $\sum_b \zeta_{bb}^\alpha$ been nonzero, $\hat{\pi}_\alpha$ would have needed an explicit symmetrization; the antisymmetry of ζ spares us.

B.2 The nine terms

Substitute $\hat{A}_\alpha = \hat{J}_\alpha - \hat{L}_\alpha - \hat{\pi}_\alpha$ into $\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta$ and multiply out, *keeping the operator order* (the factors need not commute):

$$\begin{aligned}
\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta = {} & \underbrace{\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta}_{\text{(T1)}} + \underbrace{\hat{L}_\alpha\mu_{\alpha\beta}\hat{L}_\beta}_{\text{(T2)}} + \underbrace{\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta}_{\text{(T3)}} \\
& - \underbrace{\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{L}_\beta + \hat{L}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)}_{\text{(T4) rotation–electron}} - \underbrace{\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)}_{\text{(T5) rotation–vibration}} \\
& + \underbrace{\big(\hat{L}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{L}_\beta\big)}_{\text{(T6) vibration–electron}}.
\end{aligned} \tag{B.7}$$

The three diagonal terms (T1)–(T3) come with the factor $\frac{1}{2}$ of (B.1); each cross-coupling (T4)–(T6) collects the two equal classical contributions $X\mu Y$ and $Y\mu X$, which is why it appears as a symmetric *sum* of orderings. We treat them one at a time.

B.3 Diagonal terms

(T1) $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$**.** *Is it Hermitian?* Apply Rule 1 (μ real, $\hat{J}$ Hermitian), then relabel the dummy indices $\alpha \leftrightarrow \beta$, then use $\mu_{\beta\alpha} = \mu_{\alpha\beta}$:

$$\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)^\dagger = \hat{J}_\beta\,\mu_{\alpha\beta}\,\hat{J}_\alpha = \hat{J}_\alpha\,\mu_{\beta\alpha}\,\hat{J}_\beta = \hat{J}_\alpha\,\mu_{\alpha\beta}\,\hat{J}_\beta.$$

So (T1) is Hermitian exactly as written. *Ordering / extra term?* Since μ is a function of Q only, it commutes with $\hat{J}$ [(B.3)]: $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta = \mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$. Split $\hat{J}_\alpha\hat{J}_\beta = \frac{1}{2}\{\hat{J}_\alpha, \hat{J}_\beta\} + \frac{1}{2}[\hat{J}_\alpha, \hat{J}_\beta]$ and use (B.2):

$$\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta = \tfrac{1}{2}\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} - \tfrac{i\hbar}{2}\,\mu_{\alpha\beta}\,\epsilon_{\alpha\beta\gamma}\,\hat{J}_\gamma.$$

The commutator term dies because $\mu_{\alpha\beta}$ is symmetric while $\epsilon_{\alpha\beta\gamma}$ is antisymmetric in $\alpha\beta$, so the contraction $\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$. Hence

$$\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta = \tfrac{1}{2}\,\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} \qquad \text{(Hermitian; no } c\text{-number).} \tag{B.8}$$

This is the asymmetric-top kinetic operator with Q-dependent inertia. It contains no $\hat{P}$, hence no differential-operator subtlety.

(T2) $\hat{L}_\alpha\mu_{\alpha\beta}\hat{L}_\beta$**.** Word-for-word the same as (T1) with $\hat{J} \to \hat{L}$: μ commutes with $\hat{L}$ [(B.3)], and although the $\hat{L}$ algebra carries the opposite sign, the commutator term again vanishes against symmetric μ. Thus

$$\hat{L}_\alpha \mu_{\alpha\beta} \hat{L}_\beta = \tfrac{1}{2}\,\mu_{\alpha\beta}\{\hat{L}_\alpha, \hat{L}_\beta\} \qquad \text{(Hermitian; no } c\text{-number).} \tag{B.9}$$

(T3) $\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta$**.** *Is it Hermitian?* Use $\hat{\pi}^\dagger = \hat{\pi}$ from (B.6), then relabel $\alpha \leftrightarrow \beta$ and use μ symmetric:

$$\big(\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta\big)^\dagger = \hat{\pi}_\beta\, \mu_{\alpha\beta}\, \hat{\pi}_\alpha = \hat{\pi}_\alpha\, \mu_{\beta\alpha}\, \hat{\pi}_\beta = \hat{\pi}_\alpha\, \mu_{\alpha\beta}\, \hat{\pi}_\beta.$$

So (T3) is Hermitian as written — but *only* because $\hat{\pi}$ is Hermitian, i.e. ultimately because ζ is antisymmetric (B.5). *Differential-operator form.* Write $\hat{\pi}_\alpha = \sum_b \sigma_{b\alpha} \hat{P}_b$ and slide the left momentum $\hat{P}_b$ rightward past the Q-function $\mu_{\alpha\beta}\sigma_{c\beta}$ with Rule 2 and (B.4):

$$\begin{aligned}
\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta &= \sum_{bc} \sigma_{b\alpha}\, \hat{P}_b\, \big(\mu_{\alpha\beta}\sigma_{c\beta}\big)\, \hat{P}_c = \sum_{bc} \sigma_{b\alpha} \Big[\big(\mu_{\alpha\beta}\sigma_{c\beta}\big) \hat{P}_b - i\hbar\, \partial_b\big(\mu_{\alpha\beta}\sigma_{c\beta}\big)\Big] \hat{P}_c \\
&= \underbrace{\sum_{bc} \big(\boldsymbol{\sigma\mu\sigma}^\mathsf{T}\big)_{bc}\, \hat{P}_b \hat{P}_c}_{\text{2nd order in } \hat{P}} \; \underbrace{- \, i\hbar \sum_{bc} \sigma_{b\alpha}\, \partial_b\big(\mu_{\alpha\beta}\sigma_{c\beta}\big)\, \hat{P}_c}_{\text{1st order in } \hat{P}},
\end{aligned} \tag{B.10}$$

with $(\boldsymbol{\sigma\mu\sigma}^\mathsf{T})_{bc} = \sum_{\alpha\beta} \sigma_{b\alpha}\mu_{\alpha\beta}\sigma_{c\beta}$, the Coriolis-dressed vibrational inertia. The two pieces together *are* the Hermitian operator (T3); note that every surviving term still carries at least one $\hat{P}$, so there is *no* pure constant (c-number).

B.4 Cross terms

(T4) rotation–electron, $\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta + \hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$**.** Here all three factors mutually commute: $[\hat{J}, \hat{L}] = 0$, and μ commutes with both [(B.2)–(B.3)]. A single ordering is already Hermitian,

$$\big(\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta\big)^\dagger = \hat{L}_\beta \mu_{\alpha\beta} \hat{J}_\alpha = \mu_{\alpha\beta} \hat{L}_\beta \hat{J}_\alpha = \mu_{\alpha\beta} \hat{J}_\alpha \hat{L}_\beta = \hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta,$$

and the two orderings in the sum are equal ($\hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta \overset{\alpha\leftrightarrow\beta}{=} \hat{L}_\beta \mu_{\alpha\beta} \hat{J}_\alpha = \hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta$), so

$$\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta + \hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = 2\,\mu_{\alpha\beta}\, \hat{J}_\alpha \hat{L}_\beta \qquad \text{(Hermitian; no } c\text{-number).} \tag{B.11}$$

This is the rotation–electron Coriolis coupling. No $\hat{P}$ appears, so there is no differential subtlety.

(T5) rotation–vibration, $\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$**.** Now the subtlety: $\hat{\pi}$ does *not* commute with μ,

$$[\hat{\pi}_\beta, \mu_{\alpha\beta}] = \sum_b \sigma_{b\beta} [\hat{P}_b, \mu_{\alpha\beta}] = -i\hbar \sum_b \sigma_{b\beta} \partial_b \mu_{\alpha\beta} \neq 0,$$

so the single ordering $\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta$ is *not* Hermitian by itself. Its conjugate is, however, exactly the *other* ordering (use $\hat{J}$ commutes with $\mu, \hat{\pi}$, then relabel $\alpha \leftrightarrow \beta$):

$$\big(\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta\big)^\dagger = \hat{\pi}_\beta \mu_{\alpha\beta} \hat{J}_\alpha \overset{\alpha\leftrightarrow\beta}{=} \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta.$$

The cross term is therefore of the form $\hat{X} + \hat{X}^\dagger$, which is automatically Hermitian. *This is why the classical product* $-2\,J\mu\pi$ *must be quantized as the symmetric sum, not as one* *ordering.* Pulling $\hat{J}$ out (it commutes with both μ and $\hat{\pi}$),

$$\hat{J}_\alpha \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta}\hat{J}_\beta = \hat{J}_\alpha \{\mu_{\alpha\beta}, \hat{\pi}_\beta\}, \qquad \{\mu_{\alpha\beta}, \hat{\pi}_\beta\} \equiv \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\beta \mu_{\alpha\beta}, \tag{B.12}$$

where $\{\mu_{\alpha\beta}, \hat{\pi}_\beta\}$, being the anticommutator of two Hermitian operators, is manifestly Hermitian. *Differential-operator form.* Expand the anticommutator with $\hat{\pi}_\beta = \sum_c \sigma_{c\beta}\hat{P}_c$, sliding $\hat{P}_c$ rightward in the second piece [(B.4)]:

$$\{\mu_{\alpha\beta}, \hat{\pi}_\beta\} = \sum_c \Big(\mu_{\alpha\beta}\sigma_{c\beta}\hat{P}_c + \sigma_{c\beta}\hat{P}_c\,\mu_{\alpha\beta}\Big) = \sum_c \Big(\underbrace{2\,\mu_{\alpha\beta}\sigma_{c\beta}\hat{P}_c}_{\text{1st order in } \hat{P}} \; \underbrace{-\,i\hbar\,\sigma_{c\beta}\partial_c\mu_{\alpha\beta}}_{\text{0th order in } \hat{P}}\Big). \tag{B.13}$$

Multiplied by $\hat{J}_\alpha$, the first piece is the rotation–vibration Coriolis coupling ($\hat{J}$ times $\hat{P}$); the second is the imaginary, $\hat{J}$-linear term required to make the sum Hermitian (it is not a stand-alone c-number: it still carries $\hat{J}_\alpha$). Indeed one verifies Hermiticity using (B.5): $\big(2\mu_{\alpha\beta}\sigma_{c\beta}\hat{P}_c\big)^\dagger = 2\mu_{\alpha\beta}\sigma_{c\beta}\hat{P}_c - 2i\hbar\,\partial_c(\mu_{\alpha\beta}\sigma_{c\beta})$ and $\sum_c \partial_c\sigma_{c\beta} = 0$, so the extra $-i\hbar\sigma_{c\beta}\partial_c\mu_{\alpha\beta}$ is exactly what (B.13) supplies.

(T6) vibration–electron, $\hat{L}_\alpha \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta}\hat{L}_\beta$. Identical to (T5) with $\hat{J} \to \hat{L}$ ($\hat{L}$ also commutes with μ and $\hat{\pi}$):

$$\hat{L}_\alpha \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta}\hat{L}_\beta = \hat{L}_\alpha \{\mu_{\alpha\beta}, \hat{\pi}_\beta\}, \tag{B.14}$$

with the same expansion (B.13). This is the vibration–electron angular coupling.

B.5 Assembly, and where the pseudopotential is (and is not)

Collecting (B.8)–(B.14), the quantized kernel $\frac{1}{2}\hat{A}_\alpha \mu_{\alpha\beta}\hat{A}_\beta$ is Hermitian *exactly* as written in (4.10)/(4.14), with μ sandwiched between the two factors. Two lessons stand out.

- Terms made only of *mutually commuting*, individually Hermitian operators — (T1) $\hat{J}\mu\hat{J}$, (T2) $\hat{L}\mu\hat{L}$, and (T4) $\hat{J}\mu\hat{L}$ — are Hermitian in a *single* ordering; the asymmetric-top operators (T1),(T2) shed their commutator pieces against symmetric μ.
- Terms pairing $\hat{\pi}$ with $\hat{J}$ or $\hat{L}$ — (T5),(T6) — are Hermitian only as the *symmetric sum* (anticommutator), because $\hat{\pi}$ and μ do not commute. Term (T3) $\hat{\pi}\mu\hat{\pi}$ is the fortunate middle case: with μ between two *equal* Hermitian factors it is Hermitian automatically.

No c-number from the kernel. In the explicit differential forms (B.10) and (B.13), every surviving term carries at least one momentum ($\hat{P}$, $\hat{J}$, or $\hat{L}$): the ordering of the kernel produces only second- and first-order operators, never a pure constant. *Therefore the kernel ordering does not generate the Watson pseudopotential.* The pseudopotential $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$ of (4.11) is a separate, $\mathcal{O}(\hbar^2)$ pure c-number that originates in the *curved integration measure* $\sqrt{|g|}$ — the Podolsky / Laplace–Beltrami construction — and is extracted in § 4.4 by the rescaling $\psi_W = \gamma^{1/4}\psi$. The division of labour is clean: *ordering of the kernel* → the Hermitian operators of this appendix; *the measure* → the pseudopotential of § 4.4.

Appendix C: Podolsky quantization from the ro-vibrational metric, without the anomalous angular-momentum algebra

Section 4 and Appendix B quantize the kernel of (3.22) in terms of the abstract body-fixed momenta $\hat{J}_\alpha, \hat{L}_{\mathrm{elec},\alpha}, \hat{\pi}_\alpha, \hat{P}_b$, and the argument that the rotational ordering produces no c-number [(4.10g), (B.8)] leans on the *anomalous* commutator $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\,\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ of (4.7). This appendix redoes the same quantization from the opposite end: we never introduce $\hat{J}_\alpha$ as a primitive object and we never use any commutator of body-fixed angular momenta. Instead we build the configuration-space metric g_{AB} *directly in the ro-vibrational coordinates*

$$q^A = \big(\underbrace{\phi, \theta, \chi}_{\text{Euler angles}}\ ,\ \underbrace{Q_1, \ldots, Q_f}_{\text{normal coords}}\big), \qquad f \equiv 3N_{\mathrm{nucl}} - 6, \tag{C.1}$$

and apply the Podolsky (Laplace–Beltrami) prescription (4.3) as a pure differential operator, writing out *every* partial derivative $\partial_\phi, \partial_\theta, \partial_\chi, \partial_{Q_b}$ explicitly. The electrons enter as a flat Cartesian spectator sector: $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ on $L^2(\mathbb{R}^{3N_{\mathrm{elec}}}, \prod_i d^3\bar{\mathbf{r}}_i)$, and the electronic angular momentum $\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ acts only on those coordinates, so it enters the ro-vibrational operator merely as the coordinate-independent shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\mathrm{elec}}$ of (3.19) and never touches the metric below. Everything nontrivial lives in the $(3+f)\times(3+f)$ metric on (C.1).

The output is identical to (4.14): the same Hermitian kernel and the same Watson pseudopotential $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$. The point of the exercise is that *none* of it requires the anomalous algebra — the Laplace–Beltrami operator is Hermitian by construction, and the only c-number comes from differentiating the volume element, a purely coordinate computation.

C.1 The zyz body frame and every derivative of the Euler angles

Fix the body-fixed orientation by the zyz convention

$$\mathbf{S}(\phi,\theta,\chi) = R_z(\phi)\,R_y(\theta)\,R_z(\chi), \qquad R_z(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix},\ R_y(\vartheta) = \begin{pmatrix} \cos\vartheta & 0 & \sin\vartheta \\ 0 & 1 & 0 \\ -\sin\vartheta & 0 & \cos\vartheta \end{pmatrix}, \tag{C.2}$$

so that $\mathbf{R}'_j = \mathbf{S}\bar{\mathbf{R}}_j$ as in (3.1). The body-fixed angular velocity is defined by $(\mathbf{S}^T\dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\omega_\gamma$ [eq. (3.2)]. Writing $G_z = \partial_\psi R_z|_0$ and $G_y = \partial_\vartheta R_y|_0$ for the $SO(3)$ generators,

$$G_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \tag{C.3}$$

one differentiates (C.2) term by term, using $\partial_\phi\mathbf{S} = G_z\mathbf{S}$, $\partial_\theta\mathbf{S} = R_z(\phi)G_yR_y(\theta)R_z(\chi)$, and $\partial_\chi\mathbf{S} = \mathbf{S}\,G_z$. Hence

$$\mathbf{S}^T\dot{\mathbf{S}} = \dot\phi\,\mathbf{S}^TG_z\mathbf{S} + \dot\theta\,R_z(\chi)^TG_yR_z(\chi) + \dot\chi\,G_z = \big[\dot\phi\,\mathbf{S}^T\hat{\mathbf{z}} + \dot\theta\,R_z(\chi)^T\hat{\mathbf{y}} + \dot\chi\,\hat{\mathbf{z}}\big]_\times, \tag{C.4}$$

using $R^T[\mathbf{a}]_\times R = [R^T\mathbf{a}]_\times$ and $R_y(\theta)^T\hat{\mathbf{y}} = \hat{\mathbf{y}}$. Evaluating the three body-frame vectors, $\mathbf{S}^T\hat{\mathbf{z}} = (-\sin\theta\cos\chi,\ \sin\theta\sin\chi,\ \cos\theta)$, $R_z(\chi)^T\hat{\mathbf{y}} = (\sin\chi,\ \cos\chi,\ 0)$, gives $\boldsymbol{\omega} = \mathsf{E}(\Omega)\,\dot{\boldsymbol{\Omega}}$ with

$\dot{\boldsymbol{\Omega}} = (\dot\phi, \dot\theta, \dot\chi)^T$ and

$$\boxed{\mathsf{E}(\Omega) = \begin{pmatrix} -\sin\theta\cos\chi & \sin\chi & 0 \\ \sin\theta\sin\chi & \cos\chi & 0 \\ \cos\theta & 0 & 1 \end{pmatrix}, \qquad \begin{aligned} \omega_1 &= -\dot\phi\,\sin\theta\cos\chi + \dot\theta\,\sin\chi, \\ \omega_2 &= \quad\dot\phi\,\sin\theta\sin\chi + \dot\theta\,\cos\chi, \\ \omega_3 &= \quad\dot\phi\,\cos\theta + \dot\chi. \end{aligned}} \tag{C.5}$$

Expanding along the third column (which is $\hat{\mathbf{z}}$) gives the Jacobian of the body-frame parametrization,

$$\det \mathsf{E}(\Omega) = 1{\cdot}\det\begin{pmatrix} -\sin\theta\cos\chi & \sin\chi \\ \sin\theta\sin\chi & \cos\chi \end{pmatrix} = -\sin\theta(\cos^2\chi{+}\sin^2\chi) = -\sin\theta, \qquad |\det\mathsf{E}| = \sin\theta. \tag{C.6}$$

Inverting the linear system (C.5) for $(\dot\phi, \dot\theta, \dot\chi)$ — multiply the ω_1, ω_2 equations by $(\sin\chi, \cos\chi)$ and by $(\cos\chi, -\sin\chi)$ and add — yields $\dot{\boldsymbol{\Omega}} = \mathsf{E}^{-1}\boldsymbol{\omega}$ with

$$\boxed{\mathsf{E}^{-1}(\Omega) = \begin{pmatrix} -\dfrac{\cos\chi}{\sin\theta} & \dfrac{\sin\chi}{\sin\theta} & 0 \\ \sin\chi & \cos\chi & 0 \\ \dfrac{\cos\theta\cos\chi}{\sin\theta} & -\dfrac{\cos\theta\sin\chi}{\sin\theta} & 1 \end{pmatrix}, \qquad \begin{aligned} \dot\phi &= \tfrac{1}{\sin\theta}\big(-\cos\chi\,\omega_1 + \sin\chi\,\omega_2\big), \\ \dot\theta &= \sin\chi\,\omega_1 + \cos\chi\,\omega_2, \\ \dot\chi &= \omega_3 - \tfrac{\cos\theta}{\sin\theta}\big(\sin\chi\,\omega_2 - \cos\chi\,\omega_1\big). \end{aligned}} \tag{C.7}$$

The $1/\sin\theta$ entries are the coordinate singularity at the poles $\theta = 0, \pi$; they are exactly compensated by the vanishing of $|\det\mathsf{E}| = \sin\theta$ in the volume element, so no observable is singular. Equations (C.5)–(C.7) are the *only* facts about the Euler angles we shall use; in particular we never assign a commutator to any combination of $\partial_\phi, \partial_\theta, \partial_\chi$.

C.2 The ro-vibrational metric tensor

The classical rotation–vibration kinetic energy carried into the body frame is the nuclear piece of the Routhian (3.13),

$$2T_{\rm rv} = \boldsymbol{\omega}^T\mathbf{I}_N(Q)\,\boldsymbol{\omega} + 2\,\boldsymbol{\omega}^T\sum_b \boldsymbol{\sigma}_b(Q)\,\dot Q_b + \sum_b \dot Q_b^2, \tag{C.8}$$

where $\mathbf{I}_N = \sum_j M_j(\bar R_j^2\mathbb{1} - \bar{\mathbf{R}}_j\bar{\mathbf{R}}_j^T)$ [(3.9)], the Coriolis vectors are $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab}Q_a$ with constant $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja}\times\mathbf{l}_{jb}$ [(3.17a)], and the vibrational metric is the identity because the Q_b are mass-weighted rectilinear normal coordinates, $\sum_j M_j(\partial\bar{\mathbf{R}}_j/\partial Q_a)\cdot(\partial\bar{\mathbf{R}}_j/\partial Q_b) = \delta_{ab}$ [(3.17)]. Collecting $\boldsymbol{\sigma}$ into the $3\times f$ matrix $\boldsymbol{\sigma}_{\alpha b} \equiv (\boldsymbol{\sigma}_b)_\alpha$, the velocity quadratic form is

$$2T_{\rm rv} = \begin{pmatrix} \boldsymbol{\omega} \\ \dot{\mathbf{Q}} \end{pmatrix}^T \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T & \mathbb{1}_f \end{pmatrix} \begin{pmatrix} \boldsymbol{\omega} \\ \dot{\mathbf{Q}} \end{pmatrix}. \tag{C.9}$$

Now trade the physical velocities for the coordinate velocities using the *explicit* $\boldsymbol{\omega} = \mathsf{E}(\Omega)\dot{\boldsymbol{\Omega}}$ of (C.5); the $\dot Q_b$ are already coordinate velocities. With $\left(\begin{smallmatrix}\boldsymbol{\omega}\\ \dot{\mathbf{Q}}\end{smallmatrix}\right) = \mathsf{A}\,\dot q$, $\mathsf{A} = \mathrm{diag}(\mathsf{E}, \mathbb{1}_f)$, the metric g_{AB} defined by $2T_{\rm rv} = g_{AB}\dot q^A \dot q^B$ is $g = \mathsf{A}^T(\cdots)\mathsf{A}$:

$$\boxed{g_{AB} = \begin{pmatrix} \mathsf{E}^T \mathbf{I}_N \mathsf{E} & \mathsf{E}^T\boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T\mathsf{E} & \mathbb{1}_f \end{pmatrix} = \begin{pmatrix} g_{\Omega\Omega} & g_{\Omega Q} \\ g_{Q\Omega} & g_{QQ} \end{pmatrix}, \qquad \begin{aligned} g_{\Omega\Omega} &= \mathsf{E}^T\mathbf{I}_N\mathsf{E} \quad (3\times 3), \\ g_{\Omega Q} &= \mathsf{E}^T\boldsymbol{\sigma} \quad (3\times f), \\ g_{QQ} &= \mathbb{1}_f. \end{aligned}} \tag{C.10}$$

The Euler–angle block $g_{\Omega\Omega} = \mathsf{E}^T\mathbf{I}_N\mathsf{E}$ is the inertia tensor pulled back through the kinematic matrix; the rotation–vibration block $g_{\Omega Q} = \mathsf{E}^T\boldsymbol{\sigma}$ is the Coriolis coupling; the vibrational block is flat. This is the entire geometric content of (3.22): the rotational kernel $\frac{1}{2}\mathbf{J}^T\boldsymbol{\mu}\mathbf{J}$, the Coriolis coupling $-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}$, the vibrational $\frac{1}{2}\sum_b P_b^2$, and the internal $\frac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}$ are *not* independent postulates — they are the four blocks of g^{AB} that we compute next.

C.3 Determinant and the Riemannian volume element

Take the determinant of (C.10) by the Schur complement of the identity block $g_{QQ} = \mathbb{1}_f$:

$$\det g_{AB} = \det(\mathbb{1}_f)\,\det\!\big(\mathsf{E}^T\mathbf{I}_N\mathsf{E} - \mathsf{E}^T\boldsymbol{\sigma}\,\mathbb{1}_f^{-1}\boldsymbol{\sigma}^T\mathsf{E}\big) = \det\!\big[\mathsf{E}^T(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\mathsf{E}\big]. \tag{C.11}$$

Using $\det(\mathsf{E}^T X\mathsf{E}) = (\det\mathsf{E})^2\det X$ and (C.6),

$$\boxed{\det g_{AB} = (\det\mathsf{E})^2\,\det\mathbf{I}' = \sin^2\theta\,\det\mathbf{I}'(Q), \qquad \mathbf{I}' \equiv \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T, \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^T = \sum_b \boldsymbol{\sigma}_b\boldsymbol{\sigma}_b^T,} \tag{C.12}$$

with $\mathbf{I}'$ the Watson modified inertia of (3.20). Writing $\gamma(Q) \equiv \det\mathbf{I}'(Q)$, the Riemannian volume element is

$$\boxed{\sqrt{|g|}\,d^{3+f}q = \underbrace{\sin\theta\,d\phi\,d\theta\,d\chi}_{\text{Haar on }SO(3)}\;\underbrace{\sqrt{\gamma(Q)}\,\textstyle\prod_b dQ_b}_{\rm rot.-vib.\ Jacobian}, \qquad \sqrt{|g|} = \sin\theta\,\sqrt{\gamma(Q)}.} \tag{C.13}$$

This reproduces (4.4) exactly, but now the two factors have a transparent origin: the $\sin\theta$ is $|\det\mathsf{E}|$ from the Euler-angle kinematics (C.6), and the $\sqrt{\gamma}$ is $|\det\mathsf{E}|^{-1}$ times the full determinant, i.e. the Coriolis-reduced inertia determinant. The full electronic Cartesian block multiplies (C.13) by the flat, constant $\prod_i d^3\bar{\mathbf{r}}_i$ and contributes nothing to $\sqrt{|g|}$.

C.4 The inverse metric

Invert (C.10) block-wise. With $g_{QQ} = \mathbb{1}_f$ the Schur complement of the vibrational block is $S = g_{\Omega\Omega} - g_{\Omega Q}g_{Q\Omega} = \mathsf{E}^T\mathbf{I}'\mathsf{E}$, whose inverse is $S^{-1} = \mathsf{E}^{-1}\boldsymbol{\mu}\,\mathsf{E}^{-T}$ with the Watson reciprocal inertia $\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1}$ [eq. (3.20)] and $\mathsf{E}^{-T} \equiv (\mathsf{E}^{-1})^T$. The standard block-inverse formulas give

$$\boxed{g^{AB} = \begin{pmatrix} \mathsf{E}^{-1}\boldsymbol{\mu}\,\mathsf{E}^{-T} & -\mathsf{E}^{-1}\boldsymbol{\mu}\,\boldsymbol{\sigma} \\ -\boldsymbol{\sigma}^T\boldsymbol{\mu}\,\mathsf{E}^{-T} & \mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu}\,\boldsymbol{\sigma} \end{pmatrix} = \begin{pmatrix} g^{\Omega\Omega} & g^{\Omega Q} \\ g^{Q\Omega} & g^{QQ} \end{pmatrix}.} \tag{C.14}$$

Two checks. *(i)* $\det g^{AB} = 1/\det g_{AB} = 1/(\sin^2\theta\,\gamma)$, consistent with (C.12). *(ii)* In the body-frame momentum basis (J_α, P_b) — reached by the constant-per-point linear map $\hat J_\alpha =$

$(\mathsf{E}^{-T})_{\alpha s}\hat{p}_{\Omega_s}$, $\hat{P}_b = \hat{p}_{Q_b}$ — the $\mathsf{E}^{\pm 1}$ factors are absorbed and (C.14) collapses to

$$\begin{pmatrix} \boldsymbol{\mu} & -\boldsymbol{\mu\sigma} \\ -\boldsymbol{\sigma}^T\boldsymbol{\mu} & \mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu\sigma} \end{pmatrix}, \tag{C.15}$$

which is precisely the momentum-space contravariant metric (4.10e). Equation (C.14) is the same object written in the raw coordinates q^A, with all the Euler-angle dependence made explicit through E^{-1}.

C.5 The natural first-order operators (defined, not postulated)

When the rotational block of the Laplace–Beltrami operator is written out, the combination $(\mathsf{E}^{-1})_{s\alpha}\partial_{\Omega_s}$ appears repeatedly. It is convenient to name it

$$\hat{J}_\alpha \equiv -i\hbar \sum_{s\in\{\phi,\theta,\chi\}} (\mathsf{E}^{-1})_{s\alpha}\,\partial_{\Omega_s}, \tag{C.16}$$

which, inserting the explicit E^{-1} of (C.7), reads

$$\boxed{\begin{aligned} \hat{J}_1 &= -i\hbar\Big[-\tfrac{\cos\chi}{\sin\theta}\,\partial_\phi + \sin\chi\,\partial_\theta + \tfrac{\cos\theta\cos\chi}{\sin\theta}\,\partial_\chi\Big], \\ \hat{J}_2 &= -i\hbar\Big[\ \ \tfrac{\sin\chi}{\sin\theta}\,\partial_\phi + \cos\chi\,\partial_\theta - \tfrac{\cos\theta\sin\chi}{\sin\theta}\,\partial_\chi\Big], \\ \hat{J}_3 &= -i\hbar\,\partial_\chi. \end{aligned}} \tag{C.17}$$

We stress the logic: (C.17) is a *definition*, a shorthand for the three explicit Euler-angle derivative combinations. We do *not* compute $[\hat{J}_\alpha, \hat{J}_\beta]$, we do not invoke the anomalous algebra (4.7), and nothing below depends on it. The Laplace–Beltrami operator is Hermitian because it is built from the real symmetric metric g^{AB} and the measure $\sqrt{|g|}$ — Hermiticity is automatic, not a property to be rescued by a commutator identity.

C.6 Term-by-term Podolsky quantization of (3.22)

The Podolsky prescription (4.3) turns the classical kinetic energy $T = \frac{1}{2}g^{AB}\pi_A\pi_B$ into the Hermitian operator

$$\hat{T} = -\frac{\hbar^2}{2}\frac{1}{\sqrt{|g|}}\sum_{A,B}\partial_A\big(\sqrt{|g|}\,g^{AB}\,\partial_B\big), \qquad \sqrt{|g|} = \sin\theta\,\sqrt{\gamma(Q)}. \tag{C.18}$$

We now go through the pieces of (3.22) one at a time, each quantized by (C.18), writing every derivative.

Term 0 — electrons and potential (flat / multiplicative). The electronic block of g^{AB} is flat and constant [$\mathsf{D}_{(i\alpha)(j\beta)}$ of (4.3a) is Q-, Ω-independent], so $\sqrt{|g|}$ does not depend on $\bar{\mathbf{r}}_i$ and (C.18) reduces on that sector to the ordinary Laplacian:

$$\hat{T}_e = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2, \qquad \hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}, \tag{C.19}$$

identical to (4.12). The Coulomb potential $V(Q,\{\bar{\mathbf{r}}_i\})$ of (3.8) is a real multiplication operator, unchanged. The electronic angular-momentum shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\text{elec}}$ acts across sectors that commute, so wherever $\hat{J}_\alpha$ appears below one simply replaces it by $\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$ at the end; the metric, the measure and the pseudopotential are untouched because they depend on (Ω, Q) only.

Term 1 — rotational kernel $\frac{1}{2}\mathbf{J}^T\boldsymbol{\mu}\mathbf{J}$ (the $g^{\Omega\Omega}$ block). Collect the terms of (C.18) with $A, B \in \{\phi, \theta, \chi\}$. Because $g^{\Omega\Omega} = \mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T}$ and $\boldsymbol{\mu} = \boldsymbol{\mu}(Q)$, $\gamma = \gamma(Q)$ are *independent of the Euler angles*, the factor $\sqrt{\gamma}$ in $\sqrt{|g|}$ passes through the Euler derivatives untouched and cancels against the $1/\sqrt{|g|}$ prefactor:

$$\hat{T}_{\text{rot}} = -\frac{\hbar^2}{2}\frac{1}{\sin\theta\sqrt{\gamma}}\sum_{s,t}\partial_{\Omega_s}\Big(\sin\theta\sqrt{\gamma}\,(\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T})_{st}\,\partial_{\Omega_t}\Big) = -\frac{\hbar^2}{2}\frac{1}{\sin\theta}\sum_{s,t}\partial_{\Omega_s}\Big(\sin\theta\,(\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T})_{st}\,\partial_{\Omega_t}\Big). \tag{C.20}$$

Insert $(\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T})_{st} = \sum_{\alpha\beta}(\mathsf{E}^{-1})_{s\alpha}\mu_{\alpha\beta}(\mathsf{E}^{-1})_{t\beta}$ and move the Ω-constant $\mu_{\alpha\beta}$ outside the Ω-derivatives. What remains is the divergence of $\sin\theta\,(\mathsf{E}^{-1})_{s\alpha}$ paired with $(\mathsf{E}^{-1})_{t\beta}\partial_{\Omega_t}$. A direct computation from the explicit E^{-1} of (C.7) — using $\sum_s \partial_{\Omega_s}\big(\sin\theta\,(\mathsf{E}^{-1})_{s\alpha}\big) = 0$, the coordinate statement that the body-frame generators are divergence-free on the Haar measure — collapses (C.20) to the manifestly Hermitian symmetric form

$$\boxed{\hat{T}_{\text{rot}} = \tfrac{1}{2}\sum_{\alpha\beta}\mu_{\alpha\beta}(Q)\,\hat{J}_\alpha\hat{J}_\beta,} \tag{C.21}$$

with $\hat{J}_\alpha$ the differential operators (C.17). *No c-number appears.* In the algebraic route this fact required the anomalous identity $\mu_{\alpha\beta}[\hat{J}_\alpha, \hat{J}_\beta] \propto \mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$ [(4.10g), (B.8)]; here it is instead the statement that the rotational Laplace–Beltrami operator on a shape with *fixed* Q (a rigid asymmetric top) carries no extra potential — a standard, purely coordinate result, obtained by differentiating $\sin\theta\,\mathsf{E}^{-1}$ and finding the putative c-number identically zero. As a representative check, the spherical case $\mu_{\alpha\beta} = \mu\,\delta_{\alpha\beta}$ makes (C.21) equal to $\frac{\mu}{2}\hat{J}^2$ with

$$\hat{J}^2 = -\hbar^2\Big[\tfrac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \tfrac{1}{\sin^2\theta}\big(\partial_\phi^2 - 2\cos\theta\,\partial_\phi\partial_\chi + \partial_\chi^2\big)\Big], \tag{C.22}$$

the standard symmetric-top operator, which is Hermitian on $\sin\theta\,d\phi\,d\theta\,d\chi$ with no additive constant.

Term 2 — rotation–vibration Coriolis $-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}$ (the $g^{\Omega Q}$ block). Now the derivatives mix sectors. The off-diagonal contribution to (C.18) is

$$\hat{T}_{\text{Cor}} = -\frac{\hbar^2}{2}\frac{1}{\sqrt{|g|}}\Big[\sum_{s,b}\partial_{\Omega_s}\big(\sqrt{|g|}\,g^{\Omega Q}_{sb}\,\partial_{Q_b}\big) + \sum_{b,s}\partial_{Q_b}\big(\sqrt{|g|}\,g^{Q\Omega}_{bs}\,\partial_{\Omega_s}\big)\Big]. \tag{C.23}$$

Here the second group carries ∂_{Q_b}, which *does* act on $\sqrt{|g|} = \sin\theta\sqrt{\gamma}$ and on $g^{Q\Omega}_{bs} = -(\boldsymbol{\sigma}^T\boldsymbol{\mu}\,\mathsf{E}^{-T})_{bs}$ through the Q-dependence of $\boldsymbol{\sigma}$, $\boldsymbol{\mu}$ and γ. Writing $g^{\Omega Q}_{sb} = -\sum_{\alpha\beta}(\mathsf{E}^{-1})_{s\alpha}\mu_{\alpha\beta}\sigma_{\beta b}$ and using (C.16), the first group is $\hat{J}_\alpha$ acting from the left; the second group, after the product rule

$$\partial_{Q_b}\big(\sqrt{|g|}\,g^{Q\Omega}_{bs}\,\partial_{\Omega_s}\big) = \sqrt{|g|}\,g^{Q\Omega}_{bs}\,\partial_{Q_b}\partial_{\Omega_s} + \partial_{Q_b}\big(\sqrt{|g|}\,g^{Q\Omega}_{bs}\big)\,\partial_{\Omega_s}, \tag{C.24}$$

produces the $\hat{J}_\alpha$ acting from the right *plus* a first-order term in which the Q-derivative has landed on $\sqrt{|g|}\,g^{Q\Omega}$. Assembling the two groups and symmetrizing, (C.23) becomes the Hermitian anticommutator form

$$\boxed{\hat{T}_{\rm Cor} = -\tfrac{1}{2}\sum_{\alpha} \hat{J}_\alpha\,\{\mu_{\alpha\beta}, \hat{\pi}_\beta\} = -\tfrac{1}{2}\sum_{\alpha\beta b} \hat{J}_\alpha\big(\mu_{\alpha\beta}\sigma_{\beta b}\hat{P}_b + \hat{P}_b\,\sigma_{\beta b}\mu_{\alpha\beta}\big),} \tag{C.25}$$

with $\hat{P}_b = -i\hbar\,\partial_{Q_b}$ and $\hat{\pi}_\beta = \sum_b \sigma_{\beta b}\hat{P}_b$. This is term (T5) of Appendix B [eqs. (B.12)–(B.13)], now obtained without any commutator input: the symmetric ordering is forced by the two Laplace–Beltrami groups in (C.23), and the imaginary $\hat{J}$-linear remainder in (C.25) is exactly the derivative ∂_{Q_b} of $\sqrt{|g|}\,g^{Q\Omega}$ that Hermiticity of (C.18) supplies automatically.

Term 3 — vibration, $\frac{1}{2}\sum_b P_b^2 + \frac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}$ (the g^{QQ} block). The vibrational contribution is

$$\hat{T}_{\rm vib} = -\frac{\hbar^2}{2}\,\frac{1}{\sin\theta\sqrt{\gamma}}\sum_{b,c}\partial_{Q_b}\Big(\sin\theta\sqrt{\gamma}\,\big(\delta_{bc} + (\boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma})_{bc}\big)\,\partial_{Q_c}\Big). \tag{C.26}$$

The $\sin\theta$ is Q-independent and cancels. Split $g^{QQ} = \mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}$. For the identity part, expanding the product rule with $\hat{P}_b = -i\hbar\partial_{Q_b}$,

$$-\frac{\hbar^2}{2}\,\gamma^{-1/2}\sum_b \partial_{Q_b}\big(\gamma^{1/2}\,\partial_{Q_b}\psi\big) = \tfrac{1}{2}\sum_b \hat{P}_b^2\psi - \frac{\hbar^2}{2}\sum_b\big(\tfrac{1}{2}\partial_{Q_b}\ln\gamma\big)\partial_{Q_b}\psi, \tag{C.27}$$

so the naive $\frac{1}{2}\sum_b \hat{P}_b^2$ is corrected by a first-order piece proportional to $\partial_{Q_b}\ln\gamma$ — the signature that $\hat{P}_b$ alone is *not* Hermitian on the curved measure $\sqrt{\gamma}\,dQ$. The $\boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}$ part gives the Coriolis-dressed second-order operator plus its own first-order derivative remainder, exactly term (T3) of Appendix B [eq. (B.10)]. Both first-order remainders are removed — and converted into the pseudopotential c-number — by the Podolsky rescaling of the next subsection.

C.7 The measure-induced c-number: collapse to the Watson pseudopotential

Every operator ambiguity has now been reduced to the single fact that $\hat{P}_b$ is not Hermitian on $\sqrt{\gamma}\,dQ$. Cure it by the Podolsky rescaling to the flat vibrational measure: put

$$\psi = \gamma^{-1/4}\psi_W, \qquad \psi_W = \gamma^{1/4}\psi, \qquad |\psi|^2\sqrt{|g|}\,d^{\,3+f}q = |\psi_W|^2\,\sin\theta\,d\phi\,d\theta\,d\chi\,\textstyle\prod_b dQ_b, \tag{C.28}$$

i.e. $\psi_W \in L^2(\sin\theta\,d\Omega\,dQ)$, and conjugate every vibrational operator by $\gamma^{1/4}$. Only the Q-sector is rescaled; the $\sin\theta$ stays in the measure and is carried by the rotational operators (C.21), which are already Hermitian on $\sin\theta\,d\Omega$. Working the identity-block conjugation of (C.27) outward through the product rule, exactly as in (4.10i),

$$\gamma^{1/4}\Big[-\tfrac{\hbar^2}{2}\gamma^{-1/2}\partial_{Q_b}\big(\gamma^{1/2}\partial_{Q_b}\big)\Big]\gamma^{-1/4} = \tfrac{1}{2}\hat{P}_b^2 + \underbrace{\frac{\hbar^2}{8}\Big[\partial_{Q_b}^2\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\Big]}_{c\text{-number}}, \tag{C.29}$$

the two cross terms $\pm\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}\psi_W$ cancelling. Summing over b and adding the analogous c-number residues $U_{\rm Cor}$ from the $\boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}$ dressing of (C.26) and from the cross block

(C.23) gives the total extra-potential

$$\hat{U} = U_{\text{vib}} + U_{\text{Cor}}, \qquad U_{\text{vib}} = \frac{\hbar^2}{8} \sum_b \Big[\partial^2_{Q_b} \ln\gamma + \tfrac{1}{4}(\partial_{Q_b} \ln\gamma)^2\Big]. \tag{C.30}$$

It remains to evaluate (C.30) explicitly and show it collapses to the Watson trace. Every ingredient is a Q-derivative. By Jacobi's formula for $\gamma = \det \mathbf{I}'$ and the derivative-of-inverse rule,

$$\partial_{Q_b} \ln\gamma = \mathrm{tr}\big(\boldsymbol{\mu}\, \partial_{Q_b} \mathbf{I}'\big) = \sum_{\alpha\beta} \mu_{\alpha\beta}\, \partial_{Q_b} \mathbf{I}'_{\alpha\beta}, \qquad \partial_{Q_b} \boldsymbol{\mu} = -\boldsymbol{\mu}\,(\partial_{Q_b} \mathbf{I}')\,\boldsymbol{\mu}, \tag{C.31}$$

and, because $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ with $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ *linear* in Q with *constant* $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$,

$$\partial_{Q_c} \mathbf{I}'_{\alpha\beta} = -\sum_b \big(\zeta^{\alpha}_{cb}\, \sigma_{\beta b} + \sigma_{\alpha b}\, \zeta^{\beta}_{cb}\big), \qquad \zeta^{\alpha}_{cb} \equiv (\boldsymbol{\zeta}_{cb})_\alpha = \partial_{Q_c} \sigma_{\alpha b} \text{ (constant)}. \tag{C.32}$$

Thus every $\partial_{Q_b} \ln\gamma$ and $\partial_{Q_b} \boldsymbol{\mu}$ is built from the constant Coriolis matrices $\boldsymbol{\zeta}$ and the linear $\boldsymbol{\sigma}(Q)$. Substituting (C.31)–(C.32) into U_{vib} and U_{Cor}, the gradient pieces $\propto \partial_{Q_b} \mathrm{tr}\, \boldsymbol{\mu}$ generated by the two groups cancel identically, and the residual Q-local term is fixed by the Coriolis sum rules obeyed by the $\boldsymbol{\zeta}$'s (from the orthonormality $\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$, the $\zeta^{(\alpha)}$ act as $SO(3)$ generators on the mode index). The collapse is the same one carried out in § 4.4.6, and it yields

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^{3} \mu_{\alpha\alpha}(Q), \qquad \boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}.} \tag{C.33}$$

This is the Watson pseudopotential (4.11), reproduced entirely from the coordinate derivatives of the volume element $\sqrt{|g|} = \sin\theta \sqrt{\gamma}$ — Jacobi's formula, the linearity (C.32) of $\mathbf{I}'$ in Q, and the constant $\boldsymbol{\zeta}$ sum rules. **At no point did we use the anomalous commutator (4.7), or any commutator of body-fixed angular momenta.** The rotational block (C.21) contributed no c-number for the coordinate reason that a rigid rotor has no pseudopotential; the entire $\hat{U}$ came from the Q-dependence of $\gamma = \det \mathbf{I}'$.

Assembling (C.19), (C.21), (C.25), (C.26) rescaled, and (C.33), and restoring the electronic shift $\hat{J}_\alpha \to \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$, reproduces the complete quantum Hamiltonian (4.14) term for term.

C.8 Worked example: a bent triatomic

To make "every derivative" concrete, take a bent triatomic XY_2 (e.g. H_2O), a C_{2v} asymmetric top with $N_{\text{nucl}} = 3$ and $f = 3N_{\text{nucl}} - 6 = 3$ normal coordinates: the symmetric stretch Q_1 and bend Q_2 (both a_1) and the antisymmetric stretch Q_3 (b_2). Choose the body frame as the instantaneous principal-axis system, with a, b in the molecular plane and $c \perp$ to it.

Inertia and its derivatives. At equilibrium the inertia tensor is diagonal, $\mathbf{I}_N^{\text{eq}} = \mathrm{diag}(I_a^e, I_b^e, I_c^e)$, and because the equilibrium mass distribution is planar the perpendicular-axis theorem gives the exact relation

$$I_c^e = I_a^e + I_b^e. \tag{C.34}$$

The totally symmetric coordinates Q_1, Q_2 preserve the C_{2v} symmetry, so to linear order they only *scale* the diagonal moments, whereas the antisymmetric Q_3 tilts the principal axes and enters $\mathbf{I}_N$ only at second order. Hence

$$I_\alpha(Q) = I_\alpha^e + a_\alpha^1 Q_1 + a_\alpha^2 Q_2 + \mathcal{O}(Q^2), \qquad a_\alpha^b \equiv \frac{\partial I_\alpha}{\partial Q_b}\Big|_{\text{eq}}, \tag{C.35}$$

with the inertial derivatives a_α^b computable from the mode vectors $\mathbf{l}_{jb}$ via $\partial I_{\alpha\alpha}/\partial Q_b = 2\sum_j \sqrt{M_j}\,(\bar{\mathbf{R}}_j^e{\cdot}\bar{\mathbf{R}}_j^e\,\delta - \bar{R}_{j\alpha}^e\, l_{jb,\alpha})$ evaluated at equilibrium. These are exactly the derivatives that populate the rotation–vibration Jacobian $\sqrt{\gamma(Q)}$.

Coriolis coupling. The only nonzero constant Coriolis matrix in C_{2v} couples the two a_1 modes to the b_2 mode about the b axis (the axis whose rotation maps a symmetric into an antisymmetric displacement):

$$\boldsymbol{\sigma}_1 = \zeta_{13}^b\, Q_3\, \hat{\mathbf{b}}, \quad \boldsymbol{\sigma}_2 = \zeta_{23}^b\, Q_3\, \hat{\mathbf{b}}, \quad \boldsymbol{\sigma}_3 = -\big(\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2\big)\hat{\mathbf{b}}, \qquad \zeta_{ab}^b = -\zeta_{ba}^b, \tag{C.36}$$

so $\boldsymbol{\sigma}\boldsymbol{\sigma}^T = \big[(\zeta_{13}^b Q_3)^2 + (\zeta_{23}^b Q_3)^2 + (\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2)^2\big]\hat{\mathbf{b}}\hat{\mathbf{b}}^T$ corrects only the bb element of the inertia,

$$\mathbf{I}'(Q) = \operatorname{diag}\!\big(I_a(Q),\, I_b(Q){-}s(Q),\, I_c(Q)\big), \qquad s(Q) \equiv (\zeta_{13}^b Q_3)^2{+}(\zeta_{23}^b Q_3)^2{+}(\zeta_{13}^b Q_1{+}\zeta_{23}^b Q_2)^2. \tag{C.37}$$

Note $s(Q) = \mathcal{O}(Q^2)$, so it does not affect the linear-order derivatives (C.35) but does enter the pseudopotential and the exact Jacobian.

Determinant, volume element, and its explicit derivatives. With $\mathbf{I}'$ diagonal in (C.37),

$$\gamma(Q) = \det \mathbf{I}' = I_a(Q)\,\big(I_b(Q) - s(Q)\big)\, I_c(Q), \qquad \sqrt{|g|} = \sin\theta\,\sqrt{I_a\,(I_b - s)\, I_c}. \tag{C.38}$$

The vibrational Jacobian carries the explicit derivatives

$$\partial_{Q_b} \ln\gamma = \frac{a_a^b}{I_a} {+} \frac{a_b^b - \partial_{Q_b} s}{I_b - s} {+} \frac{a_c^b}{I_c}, \qquad \partial_{Q_3} s = 2\big[(\zeta_{13}^b)^2{+}(\zeta_{23}^b)^2\big] Q_3,\;\; \partial_{Q_1} s = 2\zeta_{13}^b\big(\zeta_{13}^b Q_1{+}\zeta_{23}^b Q_2\big), \text{ etc.}, \tag{C.39}$$

which are precisely the first-order remainders that appear in (C.27) and are removed by the $\gamma^{1/4}$ rescaling.

Pseudopotential. Since $\mathbf{I}'$ is diagonal, $\boldsymbol{\mu} = \operatorname{diag}(1/I_a,\, 1/(I_b - s),\, 1/I_c)$ and the trace formula (C.33) is fully explicit:

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8}\Big[\frac{1}{I_a(Q)} + \frac{1}{I_b(Q) - s(Q)} + \frac{1}{I_c(Q)}\Big].} \tag{C.40}$$

Expanding to the order at which it first varies with the coordinates, and using (C.34),

$$\hat{U}(Q) = -\frac{\hbar^2}{8}\Big(\frac{1}{I_a^e} + \frac{1}{I_b^e} + \frac{1}{I_c^e}\Big) + \frac{\hbar^2}{8}\sum_{b=1,2}\Big(\frac{a_a^b}{(I_a^e)^2} + \frac{a_b^b}{(I_b^e)^2} + \frac{a_c^b}{(I_c^e)^2}\Big)Q_b + \mathcal{O}(Q^2). \tag{C.41}$$

The constant is an inconsequential shift of the electronic–vibrational zero; the linear term is a genuine $\hbar^2$ force on the normal coordinates — the physical content of the Watson pseudopotential for this molecule — and it was produced entirely by differentiating $\sqrt{\gamma(Q)}$ through (C.39), with no reference to angular-momentum commutators. Every step from the metric (C.10) to the surface (C.41) used only $\partial_\phi, \partial_\theta, \partial_\chi$ and ∂_{Q_b}.

C.9 Summary

1. In the ro-vibrational coordinates $(\phi, \theta, \chi, Q_b)$ the configuration-space metric is the single object (C.10), $g_{\Omega\Omega} = \mathsf{E}^T \mathbf{I}_N \mathsf{E}$, $g_{\Omega Q} = \mathsf{E}^T \boldsymbol{\sigma}$, $g_{QQ} = \mathbb{1}$, built from the explicit zyz kinematic matrix $\mathsf{E}(\Omega)$ of (C.5).

2. Its determinant factorizes as $\det g_{AB} = \sin^2\theta \, \det \mathbf{I}'$ [(C.12)], giving the volume element $\sqrt{|g|} = \sin\theta\sqrt{\gamma}$; its inverse is (C.14), $g^{\Omega\Omega} = \mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T}$, $g^{QQ} = \mathbb{1} + \boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}$.

3. Applying the Podolsky operator (C.18) term by term reproduces every piece of (3.22): the rotational kernel (C.21), the Coriolis coupling (C.25), the vibrational operator (C.26), and the flat electronic kinetic energy (C.19), each with all Euler-angle and normal-mode derivatives written out explicitly.

4. The only c-number is the measure-induced Watson pseudopotential (C.33), $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$, obtained from the coordinate derivatives of $\sqrt{\gamma}$ via Jacobi's formula and the constant Coriolis sum rules.

5. The anomalous body-fixed angular-momentum algebra (4.7) is *never used*: the Laplace–Beltrami operator is Hermitian by construction, and the derivative of the volume element does all the work the commutator did in § 4.

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